# Exercise list — Topic 4: Moral hazard, credit rationing, asset substitution, debt overhang

Worked solutions

Statements are in `Listas/lista-topic4-moral-hazard-overhang.md` (and its PDF); each exercise is cited by number and printed page and is not restated. Short answers: `lista-topic4-moral-hazard-overhang-solutions-short.pdf`. Editions are the local copies: Tirole, *The Theory of Corporate Finance* (2005); Amaro de Matos, *Theoretical Foundations of Corporate Finance* (2001); Berk = Berk, DeMarzo & Harford, *Fundamentals of Corporate Finance*, 2nd ed. (2011); Grinblatt–Titman, *Financial Markets and Corporate Strategy*, 2nd ed. (2002); Ross = Ross, Westerfield, Jordan & Lamb, *Fundamentos de Administração Financeira*, 9th ed. (2013).

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## 1. DIP financing and existing creditors — Grinblatt & Titman 16.14, p. 593

**What is asked.** A single conceptual question. Chapter 11 lets a bankrupt firm borrow new money that is paid *before* the old debt. On its face this hurts the old creditors, who move down the queue. Explain the mechanism by which they can nevertheless end up better off.

**Approach.** This is the lecture's *debt overhang* problem (Myers 1977). Write every claim as a state-by-state payoff, debt $= \min\{\text{assets}, F\}$ and equity $= \max\{\text{assets} - F, 0\}$, and compare the old creditors' payoff with and without the new project. Junior money cannot be raised when the old

debt is under water, because the old creditors capture the project's value; seniority for the new money removes that obstacle. The old creditors lose priority but gain the project's cash flows, so the comparison is "smaller share of a larger pie" versus "larger share of a smaller pie".

**Notation.**

- $s$: state of the world at the payoff date; $V_s \geq 0$: value of the assets in place in state $s$.
- $F > 0$: face value of the existing debt, protected by a covenant that forbids issuing debt senior to it.
- $I > 0$: cost of a new project that must be carried out inside the firm; $X_s \geq 0$: its payoff in state $s$; risk-neutral investors, zero riskless rate (as in the book's and the lecture's examples).
- $N$: face value of the new debtor-in-possession (DIP) debt, senior to $F$; $D$, $E$: market values of the existing debt and of equity.

**Step 1 — Why the project is not funded outside bankruptcy.** We first show that, without a seniority exception, a good project is passed up; this is the status quo the old creditors compare against. Without DIP financing, new money can only be junior to $F$. In state $s$ a junior claimant receives at most what is left after the senior debt,

$$\text{junior payoff}_s \leq \max\{V_s + X_s - F,\, 0\}. \tag{1}$$

In the book's Lily Pharmaceuticals example (pp. 564–566), in the unfavourable state the assets in place are worth nothing without further development ($V_s = 0$), development costs $I = 100$ and yields $X_s = 150$, and the senior obligation is $F = 110$. Substituting into (1),

$$\begin{aligned} \text{junior payoff} &\leq \max\{(0) + (150) - (110),\, 0\} \\ &= \max\{40,\, 0\} \\ &= 40 < 100 = I, \end{aligned}$$

so no one lends: junior lenders would pay 100 for a claim worth at most 40. Yet the project is worth doing,

$$\begin{aligned} NPV &= X_s - I \\ &= (150) - (100) \\ &= 50 > 0, \end{aligned}$$

so it is passed up, the firm defaults, and the existing lenders receive $\min\{V_s, F\} = \min\{0, 110\} = 0$ in that state.

**Step 2 — DIP debt makes it fundable.** We now let the new money jump the queue and recompute the old creditors' payoff. In Chapter 11 the firm may issue debt senior to $F$. A DIP loan of $I = 100$ at the zero riskless rate has face $N = 100$ and is riskless because $150 \geq 100$. The existing lenders now receive

$$\begin{aligned} \text{existing debt payoff} &= \min\{V_s + X_s - N,\, F\} \\ &= \min\{(0) + (150) - (100),\, (110)\} \\ &= \min\{50,\, 110\} \\ &= 50, \end{aligned} \tag{2}$$

instead of 0: they gain $50 - 0 = 50$, the whole NPV, even though they lost their priority.

**Step 3 — The general argument.** The example is not special; we show the old creditors never lose when the DIP money funds a project that pays back at least its cost. With a riskless DIP loan ($N = I \leq X_s$ in every state), the existing debt's payoff with the project is $\min\{V_s + X_s - I,\, F\}$,

and without it $\min\{V_s,\, F\}$. Since $X_s - I \geq 0$, the first argument of the minimum is larger with the project, $V_s + X_s - I \geq V_s$, and $\min\{\cdot, F\}$ is nondecreasing in its first argument, so

$$\min\{V_s + X_s - I,\, F\} \geq \min\{V_s,\, F\} \qquad \text{for every } s, \tag{3}$$

with strict inequality in every state where the old debt was impaired ($V_s < F$). Seniority is diluted, but the asset pool the old creditors are paid from grows by more than the new claim.

**Step 4 — The lecture's numbers.** Topic 4B (slides 22–31): assets in place worth 500 (boom) or 200 (slump), each with probability 0.5; $F = 300$; a project costs $I = 60$ and adds 100 in both states. We value both claims without the project, then with it. Without the project,

$$\begin{aligned} D &= (0.5)\min\{500, 300\} + (0.5)\min\{200, 300\} \\ &= (0.5)(300) + (0.5)(200) = 150 + 100 = 250, \\ E &= (0.5)\max\{500 - 300, 0\} + (0.5)\max\{200 - 300, 0\} \\ &= (0.5)(200) + (0.5)(0) = 100. \end{aligned}$$

Financing the 60 with senior DIP debt ($N = 60$, riskless since the slump assets $200+100 = 300 \geq 60$), the assets become $500 + 100 = 600$ (boom) and $200 + 100 = 300$ (slump), and the old debt receives, by (2),

$$\begin{aligned} \text{boom:}\ & \min\{(600) - (60),\, (300)\} = \min\{540, 300\} = 300, \\ \text{slump:}\ & \min\{(300) - (60),\, (300)\} = \min\{240, 300\} = 240, \\ D &= (0.5)(300) + (0.5)(240) \\ &= 150 + 120 = 270, \end{aligned} \tag{4}$$

and equity receives $\max\{600 - 60 - 300, 0\} = 240$ (boom) and $\max\{300 - 60 - 300, 0\} = 0$ (slump), so $E = (0.5)(240) + (0.5)(0) = 120$. Both claims rise by 20 ($270 - 250$ and $120 - 100$), and $20 + 20 = 40 = 100 - 60$ is the project's NPV.

**Answer (16.14).** The old creditors benefit because, without senior new money, a positive-NPV project cannot be financed (debt overhang) and they are paid out of the assets in place only. With DIP financing the project is done; the DIP lender is repaid first, but the project adds more to the pool than the DIP claim takes out ($X_s \geq I$), so the old creditors' payoff rises in every impaired state (\$0 to \$50 in the Lily example; \$250 to \$270 in the lecture's example).

**Step 5 — The limit.** (3) needs $X_s \geq I$: a good, safe project. The same provision lets a distressed firm borrow to keep operating when it would be worth more liquidated; then $X_s < I$ in some state and the old creditors lose (Result 16.11).

**Intuition.** Debt overhang means the old creditors capture part of the value of new investment while the new money bears its full cost, so junior money cannot be raised. Letting new money jump the queue removes the overhang. The old creditors, who would otherwise get little or nothing, take a smaller slice of a larger pie. A covenant that blocks this hurts them, and a diffuse group of bondholders cannot easily agree to lend the money themselves (free-rider problem). Chapter 11 unlocks senior financing only in distress, which keeps most of the protection the covenant gives in normal times.

**Common mistakes.**

- Comparing the old creditors' payoff with the project under DIP financing against their payoff with the project under *junior* financing. The relevant status quo is no project at all, because junior money cannot be raised.
- Crediting the old creditors with the project's gross payoff $X_s$ and forgetting that the DIP lender is repaid $N = I$ first; their gain is bounded by $X_s - I$, the project's NPV.
- Stating that DIP financing always helps creditors. It helps only when the funded project pays back at least its cost ($X_s \geq I$); financing a value-destroying continuation hurts them.

**Ref:** Grinblatt & Titman Ch. 16, § 16.2 "Underinvestment as a Consequence of Debt Overhang" (Lily example), pp. 564–566; § 16.3 "How Chapter 11 Bankruptcy Mitigates Debt Holder–Equity Holder Incentive Problems", Result 16.11, p. 580. Lecture Topic 4B, slides 22–31.

## 2. Asset substitution: project A or B — Grinblatt & Titman 16.1, p. 591

**What is asked.** (a) The net present value of each project for the firm as a whole. (b) Which project the shareholders, who control the firm, would pick, and the reason for their choice.

**Approach.** With risk-neutral investors and a zero riskless rate, a claim's value is its expected payoff. Debt and equity split each state's payoff: debt gets $\min\{X, F\}$, equity gets $\max\{X - F, 0\}$ (limited liability). Because the cash on hand exactly equals the face value, equity is an at-the-money call option on the project payoff, so it values variance. This is the lecture's *asset substitution* (risk-shifting) problem: comparing firm value (a) with equity value (b) shows the gap between what is efficient and what shareholders choose.

**Notation.**

- $j \in \{A, B\}$: project; $X_{j,s}$: its payoff in state $s \in \{\text{fav}, \text{unf}\}$, each with probability $\pi = 0.5$.
- $C = 100$: cash on hand, all spent on the project; $F = 100$: debt due next period; $r_f = 0$; risk-neutral investors.
- $V_j = \mathbb{E}[X_j]$: firm value under $j$; $E_j$, $D_j$: values of equity and debt under $j$.

**Step 1 — (a) NPVs.** We value each project's payoff and subtract the \$100 it costs. With $r_f = 0$ and risk neutrality the present value is the expected payoff:

$$\begin{aligned} V_A &= \pi X_{A,\text{fav}} + (1-\pi) X_{A,\text{unf}} \qquad (5)\\ &= (0.5)(120) + (0.5)(60) \\ &= 60 + 30 = 90, \\ NPV_A &= V_A - C \qquad (6)\\ &= (90) - (100) \\ &= -10, \\ V_B &= \pi X_{B,\text{fav}} + (1-\pi) X_{B,\text{unf}} \qquad (7)\\ &= (0.5)(101) + (0.5)(101) \\ &= 50.5 + 50.5 = 101, \\ NPV_B &= V_B - C \\ &= (101) - (100) \\ &= 1. \end{aligned}$$

**Answer (a).** $NPV_A = -\$10$ million, $NPV_B = +\$1$ million: only B creates value.

**Step 2 — (b) Payoffs to each claim, state by state.** The shareholders care about their own claim, not about firm value, so we split each project's payoff between the claims. Equity receives $\max\{X - F, 0\}$ and debt $\min\{X, F\}$:

| | Equity | | Debt | |
|---|---|---|---|---|
| | Favourable | Unfavourable | Favourable | Unfavourable |
| A | $120 - 100 = 20$ | 0 | 100 | 60 |
| B | $101 - 100 = 1$ | $101 - 100 = 1$ | 100 | 100 |

(In A's unfavourable state $\max\{60-100,0\}=0$ and $\min\{60,100\}=60$.) Taking expectations,

$$E_A = (0.5)(20)+(0.5)(0) = 10+0 = 10, \qquad D_A = (0.5)(100)+(0.5)(60) = 50+30 = 80, \tag{8}$$
$$E_B = (0.5)(1)+(0.5)(1) = 0.5+0.5 = 1, \qquad D_B = (0.5)(100)+(0.5)(100) = 100. \tag{9}$$

Check: $E_A + D_A = 10+80 = 90 = V_A$ and $E_B + D_B = 1+100 = 101 = V_B$.

**Step 3 — (b) The choice.** We compare the shareholders' payoffs and decompose the difference into a value effect and a transfer. Equity holders, who control the firm, want project A: $E_A = 10 > E_B = 1$, although A destroys value. Writing $E_j = V_j - D_j$ and using (8)–(9),

$$\begin{aligned} E_A - E_B &= (V_A - V_B) - (D_A - D_B) \\ &= (90-101) - (80-100) \\ &= (-11) - (-20) \\ &= 9 : \end{aligned} \tag{10}$$

switching to A lowers firm value by 11 but moves 20 from the creditors to the shareholders.

**Answer (b).** The shareholders want project A ($E_A = \$10$ million $> E_B = \$1$ million), because limited liability lets them keep A's upside while the creditors bear its downside: the \$20 million transferred from debt exceeds the \$11 million of value destroyed.

**Intuition.** With \$100 due and \$100 in cash, the equity is an at-the-money call on the firm's assets. The safe project leaves it almost worthless (1); the risky one gives it the whole upside (20) while the creditors absorb the downside (60 instead of 100). This is the lecture's asset-substitution (risk-shifting) problem: limited liability makes levered equity prefer variance, even at negative NPV.

**Common mistakes.**

- Writing equity's payoff as $X - F$ in every state. In A's unfavourable state that gives $-40$; limited liability caps the loss at 0, and that cap is exactly what makes A attractive.
- Answering (b) with the NPVs of (a). Firm value and equity value rank the projects in opposite orders; (b) asks about the shareholders' claim.
- Forgetting that the \$100 of cash is spent on the project: the NPV subtracts it, but the payoff split in (b) uses only the project payoff and the \$100 face value.

**Ref:** Grinblatt & Titman Ch. 16, § 16.2 "The Asset Substitution Problem", pp. 569–572 (Results 16.7–16.8). Lecture Topic 4B, slides 2–8.

## 3. Debt overhang on vacant land — Berk 16.25, pp. 494–495

**What is asked.** (a) Today's values of equity and debt if the land stays vacant. (b) The NPV of developing the land, for the firm as a whole. (c) Today's values of equity and debt if the shareholders put in \$20 million and the land is developed. (d) Whether the shareholders gain from putting in that money, given (c).

**Approach.** All cash flows are riskless, so every claim is its payoff in one year discounted at $r_f = 10\%$; debt gets $\min\{\text{land value}, 15\}$ and equity the rest. This is the lecture's *debt overhang* (under-investment) problem: the debt is under water, so part of the value a new project creates goes to the creditors, who pay none of its cost. Comparing the NPV (b) with the shareholders' gain (d) shows the wedge.

**Notation.**

- $F = 15$: face value of the debt, due in one year; $r_f = 0.10$; all cash flows riskless; no taxes.
- $L_0 = 10$: value of the vacant land in one year; $L_1 = 35$: value of the developed land in one year; $K = 20$: development cost paid today.

- $E$, $D$: values today of equity and debt.

**Step 1 — (a) Land left vacant.** We split the land's value in one year between the claims and discount. In one year the firm is worth $10 < 15$: the creditors take 10 and the shareholders get 0. Discounting at $r_f$,

$$\begin{aligned} D &= \frac{\min\{L_0, F\}}{1+r_f} \\ &= \frac{\min\{10, 15\}}{1+0.10} \\ &= \frac{(10)}{(1.10)} \\ &= 9.0909, \\ E &= \frac{\max\{L_0 - F, 0\}}{1+r_f} = \frac{\max\{10-15, 0\}}{1.10} = \frac{(0)}{(1.10)} = 0. \end{aligned} \tag{11}$$

**Answer (a).** Equity is worth \$0; debt is worth \$9.09 million.

**Step 2 — (b) NPV of developing.** The NPV counts only the cash flows that change with the decision. Developing replaces land worth 10 by land worth 35, so the incremental cash flow in one year is $35 - 10 = 25$, against 20 paid today:

$$\begin{aligned} NPV &= -K + \frac{L_1 - L_0}{1+r_f} \\ &= -(20) + \frac{(35)-(10)}{(1.10)} \\ &= -20 + \frac{25}{1.10} \\ &= -20 + 22.7273 \\ &= 2.7273. \end{aligned} \tag{12}$$

(Ignoring the vacant-land value forgone would give $-20 + 35/1.10 = 11.8182$, the NPV measured against a firm with no land at all; the decision-relevant figure is (12).)

**Answer (b).** $NPV = \$2.73$ million $> 0$: developing creates value.

**Step 3 — (c) Values if developed with \$20 of new equity.** We repeat Step 1 with the developed land. The \$20 is spent on development, so it is not an asset in one year. In one year the firm is worth $35 \geq 15$: the debt is paid in full, $\min\{35, 15\} = 15$, and equity keeps $\max\{35 - 15, 0\} = 20$. Hence

$$\begin{aligned} D &= \frac{(15)}{(1.10)} \\ &= 13.6364, \end{aligned} \tag{13}$$

$$\begin{aligned} E &= \frac{(35)-(15)}{(1.10)} \\ &= \frac{20}{1.10} \\ &= 18.1818. \end{aligned} \tag{14}$$

**Answer (c).** Equity is worth \$18.18 million and debt \$13.64 million today.

**Step 4 — (d) Will the shareholders put up \$20?.** The shareholders compare what they hold after contributing with what they hold if they do nothing. Their gain is the equity value (14), minus

the equity value without development (Step 1), minus the contribution:

$$\begin{aligned}\Delta E &= E_{\text{dev}} - E_{\text{vacant}} - K \\ &= (18.1818) - (0) - (20) \\ &= -1.8182 < 0, \\ \Delta D &= D_{\text{dev}} - D_{\text{vacant}} \\ &= (13.6364) - (9.0909) \\ &= 4.5455.\end{aligned}$$

No: they would lose \$1.82 million. The sum $-1.8182 + 4.5455 = 2.7273$ is the NPV (12); the creditors capture more than all of it.

**Answer (d).** No. The shareholders would pay \$20 million for an equity stake worth \$18.18 million (a loss of \$1.82 million), while the debt gains \$4.55 million; the positive-NPV project is passed up.

**Intuition.** The debt is under water ($10 < 15$), so the first \$5 million of any value the project creates goes to the creditors, who pay none of its cost. The shareholders bear the \$20 million and receive only the value above the \$15 million face. This is Myers' debt overhang: leverage acts as a tax on new investment, and a positive-NPV project is passed up. Writing the debt down (renegotiation, as in the lecture) or senior new financing would fix it.

**Common mistakes.**

- Computing the NPV as $-20 + 35/1.10 = 11.82$: the vacant land worth \$10 million in one year is given up, so it is part of the project's cost.
- Answering (d) with $E_{\text{dev}} = 18.18 > 0$, "so yes". The shareholders must compare the stake with the \$20 million they put in.
- Valuing the debt in (a) at the face value $15/1.10$; the creditors can get at most what the land is worth, $\min\{10, 15\}$.

**Ref:** Berk Ch. 16, § 16.6 "Agency Costs and Information", "Equity-Debt Holder Conflicts" (under-investment), pp. 483–484. Lecture Topic 4B, slides 19–30.

## 4. Short-term debt and incentive problems — Grinblatt & Titman 16.11, p. 593

**What is asked.** A single conceptual question: why the conflicts between shareholders and creditors (asset substitution, debt overhang, under-maintenance) do less damage when the firm finances itself with debt that matures soon and must be rolled over, rather than with long-term debt.

**Approach.** The conflicts exist because equity chooses the firm's strategy *after* the terms of the debt are fixed, so it can change the value of the creditors' claim (the lecture's asset substitution and debt overhang). We write equity as firm value minus debt value and ask what happens to the debt term when the debt is repriced after the choice is made: if new lenders price the debt knowing the strategy, the debt term no longer depends on the strategy and equity internalizes the full value consequences.

**Notation.**

- $j$: investment strategy (project choice, risk level, maintenance), chosen by the equity holders after the debt is in place; $X_j \geq 0$: resulting terminal firm value, $V_j = \mathbb{E}[X_j]$ (risk neutrality, $r_f = 0$).
- $F$: face value of the debt; $L$: amount the firm owes when its short-term debt is rolled over; $D_j$, $E_j$: values of debt and equity under $j$.

**Step 1 — Where the conflict comes from.** We locate the conflict in the debt term of equity's value. With long-term debt, $F$ is fixed before $j$ is chosen, and since $\max\{X-F,0\} = X - \min\{X,F\}$ in every state,

$$\begin{aligned} E_j &= \mathbb{E}[\max\{X_j - F, 0\}] \\ &= V_j - D_j, \qquad D_j = \mathbb{E}[\min\{X_j, F\}]. \end{aligned} \tag{15}$$

Equity can gain by picking a $j$ that lowers $D_j$ (more risk, less investment, payouts) even when $V_j$ falls: comparing two strategies, $E_j - E_k = (V_j - V_k) - (D_j - D_k)$, and the second term is a transfer from creditors whenever $D_j < D_k$.

**Step 2 — Debt that is repriced when the choice is made.** We now let the debt be repriced after the choice and show the transfer term disappears. If the debt matures before or at the decision and must be rolled over, the new lenders see $j$ and set the new face $F_j$ so that their claim is worth what they lend, $\mathbb{E}[\min\{X_j, F_j\}] = L$. Substituting into (15),

$$E_j = V_j - (L), \tag{16}$$

and $L$ does not depend on $j$: equity bears the full consequences of its choice and picks the value-maximizing strategy. Short-term debt is worth close to its face whatever the firm does, so there is little value to expropriate.

**Step 3 — Illustration (Grinblatt & Titman 16.1).** Take the firm of G&T Exercise 16.1: \$100 of debt, projects A (pays 120 or 60) and B (pays 101 in both states), each state with probability 0.5, $r_f = 0$. With long-term debt, equity is worth $0.5(20) + 0.5(0) = 10$ under A and $0.5(1) + 0.5(1) = 1$ under B, so it picks the value-destroying A. Suppose instead the \$100 had to be refinanced after the project is chosen. Firm values are $V_A = 0.5(120) + 0.5(60) = 90$ and $V_B = 0.5(101) + 0.5(101) = 101$; by (16) with $L = 100$,

$$E_A = (90) - (100) = -10, \qquad E_B = (101) - (100) = 1.$$

Under A the firm cannot raise \$100 (no claim on payoffs worth 90 is worth 100), so the equity holders choose B, the positive-NPV project, instead of A.

**Step 4 — Debt overhang.** Myers (1977): if the existing debt matures before the firm must raise money for a new project, the debt is repriced together with the project decision. The project's benefit to creditors (a safer firm) then shows up as a lower borrowing rate that accrues to equity, and positive-NPV projects are taken. In Unitron's case, the riskless 12% project would lower the firm's 13% borrowing rate and become attractive.

**Step 5 — Asset substitution.** The lending rate is renegotiated after the risk choice is made and seen, so a risky choice raises the firm's own cost of funds instead of transferring wealth from existing lenders. The threat of not being rolled over also disciplines the choice (lecture Topic 4B, slide 16).

**Step 6 — The cost.** Rolling over exposes the firm to refinancing and interest-rate risk: an unexpected rise in rates can push a highly levered firm into distress (hence the advice to pair short-term debt with interest-rate hedges).

**Answer (16.11).** Short-term debt is repriced (rolled over) after the shareholders' investment and risk choices are made and observed, so its terms reflect those choices: equity bears the full value consequences, $E_j = V_j - L$, and cannot shift wealth from creditors through risk-taking or under-investment. Short-term debt is also close to riskless, so there is little value to expropriate. The price is rollover and interest-rate risk.

**Intuition.** The conflicts exist because equity can change the value of a claim whose terms are already fixed. The shorter the maturity, the sooner those terms are reset at prices that reflect what equity has done, and the less wealth equity can take from creditors. The incentive problem shrinks, and the firm takes on rollover risk in exchange.

**Common mistakes.**

- Saying short-term debt helps "because it is safer". The point is timing: the debt is priced after the decision, so the decision's effect on the creditors comes back to equity through the new rate.
- Confusing maturity with seniority. Seniority decides who is paid first in default; maturity decides when the terms are reset.
- Omitting the cost. Short maturities expose the firm to refinancing and interest-rate risk, so they are not a free cure.

**Ref:** Grinblatt & Titman Ch. 16, § 16.4 "The Use of Short-Term versus Long-Term Debt", pp. 584–585 (Result 16.12). Lecture Topic 4B, slide 16.

## 5. Random financing — Tirole 3.1, p. 144

**What is asked.** (i) Write the condition under which investors are willing to sign a contract in which the borrower contributes $\hat{A}$ of her cash, the project then goes ahead only with probability $x$, and she gets $R_b$ on success. (ii) Show that a borrower with too little cash for certain financing ($A < \bar{A}$) should put *all* her cash in, and describe how the financing probability depends on $A$.

**Approach.** This is the Holmström–Tirole fixed-investment model of the lecture (Topic 4A). The two constraints are the borrower's incentive constraint ($\text{IC}_b$), which caps the *pledgeable income* at $\mathcal{P} = p_H(R - B/\Delta p)$, and the investors' breakeven constraint ($\text{IR}_l$). Below $\bar{A}$ the project cannot be financed for sure; a lottery over financing scales the investors' outlay and the pledgeable income by $x$ but not the borrower's contribution, which convexifies the indivisible project. We write ($\text{IR}_l$) for the lottery, show ($\text{IC}_b$) must hold, and maximise the borrower's utility over $(x, \hat{A})$.

**Notation.**

- $I > 0$: fixed investment; $A \in (0, \bar{A})$: borrower's cash; $\hat{A} \in [0, A]$: the part she contributes; $x \in [0, 1]$: probability that the project is financed.
- $R > 0$: income on success (0 on failure); $R_b \geq 0$: borrower's reward on success; $R_l = R - R_b$: investors' claim on success.
- $p_H > p_L$: success probabilities when she behaves / misbehaves; $\Delta p = p_H - p_L$; $B > 0$: private benefit from misbehaving.
- $\mathcal{P} = p_H(R - B/\Delta p)$: pledgeable income; $\bar{A} = I - \mathcal{P}$: minimum cash for financing with probability 1 (the statement's definition); $U_b$: borrower's net utility. Rate of interest 0, everybody risk neutral.

**Assumption.** The borrower hands over $\hat{A}$ when the contract is signed, whether or not the lottery then selects the project (this is the reading in Tirole's answer). If $\hat{A}$ were paid only when the project goes ahead, the expected outlay would be $x(I - \hat{A})$ and the breakeven condition $x(I - \hat{A}) \leq x\mathcal{P}$ would require $\hat{A} \geq \bar{A} > A$: randomizing would achieve nothing.

**Step 1 — (i) Investors' breakeven condition.** Investors receive $\hat{A}$ for sure, pay $I$ with probability $x$, and, if the borrower behaves, collect $R_l$ with probability $p_H$ when the project runs:

$$\underbrace{\hat{A}}_{\text{received}} + \underbrace{x\,p_H(R - R_b)}_{\text{expected repayment}} \geq \underbrace{xI}_{\text{expected outlay}}$$

$$x\,p_H(R - R_b) \geq xI - \hat{A}. \tag{17}$$

**Answer (i).** $x\,p_H(R - R_b) \geq xI - \hat{A}$ (with $R_b \geq B/\Delta p$ so that she behaves).

**Step 2 — The contract must induce good behaviour.** We check that a contract under which she shirks is never worth signing, so ($\text{IC}_b$) can be imposed. Suppose $(\Delta p)R_b < B$, so she shirks.

Investors break even only if $xp_L(R-R_b) \geq xI - \hat{A}$, i.e. $-\hat{A} \leq xp_L(R-R_b) - xI$. Her net utility is then

$$\begin{aligned} x(p_L R_b + B) - \hat{A} &\leq x(p_L R_b + B) + \big(xp_L(R-R_b) - xI\big) \\ &= xp_L R_b + xB + xp_L R - xp_L R_b - xI \\ &= x(p_L R + B - I) < 0 \end{aligned} \tag{18}$$

by (3.2), worse than not signing. So the contract satisfies ($\text{IC}_b$), $(\Delta p)R_b \geq B$, i.e. $R_b \geq B/\Delta p$.

**Step 3 — Breakeven in terms of $\bar{A}$.** We combine the two constraints into one feasibility condition on $(x, \hat{A})$. ($\text{IC}_b$) gives $R - R_b \leq R - B/\Delta p$, so the left side of (17) is at most $x\,p_H(R - B/\Delta p) = x\mathcal{P}$. A contract exists for $(x, \hat{A})$ only if

$$\begin{aligned} xI - \hat{A} &\leq x\,p_H\left(R - \frac{B}{\Delta p}\right) \\ xI - \hat{A} &\leq x(\mathcal{P}) \\ x(I - \mathcal{P}) &\leq \hat{A} \\ x(\bar{A}) &\leq \hat{A} \\ x &\leq \frac{\hat{A}}{\bar{A}}, \end{aligned} \tag{19}$$

substituting $I - \mathcal{P} = \bar{A}$ from the statement's definition (third to fourth line) and dividing by $\bar{A} > 0$ (last line). Conversely, if (19) holds, $R_b = B/\Delta p$ satisfies both constraints, so (19) is necessary and sufficient.

**Step 4 — (ii) The borrower's objective.** We write her utility and use the binding breakeven constraint to express it in terms of $x$ only. She keeps $A - \hat{A}$ and, with probability $x$, earns $R_b$ with probability $p_H$; net of the $A$ she would consume anyway,

$$\begin{aligned} U_b &= x\,p_H R_b + (A - \hat{A}) - A \\ &= x\,p_H R_b - \hat{A}. \end{aligned} \tag{20}$$

Competitive investors make (17) bind; expanding and solving for $xp_H R_b$,

$$\begin{aligned} xp_H R - xp_H R_b &= xI - \hat{A} \\ xp_H R_b &= xp_H R - xI + \hat{A}. \end{aligned}$$

Substituting into (20),

$$\begin{aligned} U_b &= (xp_H R - xI + \hat{A}) - \hat{A} \\ &= x\,(p_H R - I). \end{aligned} \tag{21}$$

**Step 5 — (ii) Optimal $\hat{A}$ and $x$.** We maximise (21) over the feasible set (19). Differentiating,

$$\frac{\partial U_b}{\partial x} = p_H R - I > 0, \qquad \frac{\partial U_b}{\partial \hat{A}} = 0,$$

so $U_b$ rises with $x$ and $\hat{A}$ matters only through feasibility. By (19), the largest feasible $x$ is $\hat{A}/\bar{A}$ (below 1 because $\hat{A} \leq A < \bar{A}$), which increases with $\hat{A}$: substituting it, $U_b = (\hat{A}/\bar{A})(p_H R - I)$, increasing in $\hat{A}$ on $[0, A]$. So she contributes everything:

$$\hat{A}^* = A,$$

$$x^* = \frac{A}{\bar{A}}, \tag{22}$$

$$U_b^* = \frac{A}{\bar{A}}\,(p_H R - I). \tag{23}$$

At $x^*$ the binding (17), divided by $x^* = A/\bar{A}$, gives $p_H(R - R_b) = I - A/x^* = I - \bar{A} = \mathcal{P}$, so $R - R_b = R - B/\Delta p$ and $R_b = B/\Delta p$: ($\text{IC}_b$) binds.

**Step 6 — (ii) How $x^*$ varies with $A$.** $x^* = A/\bar{A}$ rises linearly from 0 at $A = 0$ to 1 at $A = \bar{A}$, and stays at 1 above $\bar{A}$. Utility (23) therefore rises smoothly to the full NPV instead of jumping at $\bar{A}$ as in (3.6).

*Numerical illustration.* $I = 100$, $R = 150$, $p_H = 0.8$, $p_L = 0.4$, $B = 30$, $A = 10$: (3.2) holds ($0.4(150) - 100 + 30 = -10 < 0$); $B/\Delta p = 75$; $\mathcal{P} = 0.8(150 - 75) = 60$; $\bar{A} = 100 - 60 = 40$; NPV $= 0.8(150) - 100 = 20$. Then $x^* = 10/40 = 0.25$ and $U_b^* = 0.25(20) = 5$; investors pay $0.25(100) - 10 = 15$ in expectation and receive $0.25(60) = 15$.

**Answer (ii).** $\hat{A}^* = A$; the financing probability is $x^* = A/\bar{A}$, rising linearly from 0 at $A = 0$ to 1 at $A = \bar{A}$ (and $x^* = 1$ for $A \geq \bar{A}$); her utility is $x^*(p_H R - I)$.

**Intuition.** The obstacle is not the size of the NPV but the pledgeable income per unit of investment. A poor borrower can still buy a *fraction* of the deal: each unit of her cash covers the investors' shortfall on $1/\bar{A}$ units of probability of financing. Since she captures the whole NPV whenever the project runs, she wants as high a probability as her cash can buy and puts all of it in. Lotteries convexify the indivisible project.

**Common mistakes.**

- Writing the breakeven as $xp_H(R - R_b) \geq x(I - \hat{A})$, i.e. taking her contribution only when the project goes ahead. Then $x$ cancels and randomisation achieves nothing.
- Using $R$ instead of $R - B/\Delta p$ as the investors' maximal claim: without ($\text{IC}_b$) she shirks, and by (3.2) no one breaks even.
- Claiming she keeps some cash "for safety". Cash kept aside earns nothing extra, while each unit put in raises $x$ by $1/\bar{A}$ and her utility by $(p_H R - I)/\bar{A} > 0$.

**Book answer.** Matches the book's answer (p. 571): breakeven $xI - A \leq xp_H(R - R_b)$, equivalently $x\bar{A} \leq A$; $U_b = x(p_H R - I)$, so $x^* = A/\bar{A}$, rising from 0 to 1 as $A$ goes from 0 to $\bar{A}$.

**Ref:** Tirole Ch. 3, § 3.2.1–3.2.2 "The Fixed-Investment Model", "The Lenders' Credit Analysis", pp. 115–117 (footnote 9, p. 116). Lecture Topic 4A, slides 26–34.

## 6. Zymase: leverage and strategy choice — Berk 16.26, p. 495

**What is asked.** (a) Which strategy gives the largest expected payoff to the firm as a whole. (b) and (c) Which strategy gives the shareholders the largest expected payoff when \$40 million, respectively \$110 million, of debt is due at the payoff date. (d) How much expected firm value is lost, in each case, when management follows the shareholders' preference instead of the value-maximising strategy (the expected agency cost of debt).

**Approach.** Split each strategy's payoff state by state into debt, $\min\{X, F\}$, and equity, $\max\{X - F, 0\}$, and take expectations. Equity is a call option on the firm, so as $F$ rises it favours strategies whose payoff lies above $F$ with some probability, even if their mean is lower: the lecture's *asset substitution* (excessive risk-taking). The agency cost is the drop in expected *firm* payoff caused by that choice, which is the loss the shareholders bear ex ante when creditors price the debt.

**Notation.**

- $j \in \{A, B, C\}$: research strategy; $X_j$: its after-tax payoff (\$ million), with the probabilities in the table.
- $F \in \{40, 110\}$: face value of the debt, due when the payoff arrives; equity gets $\max\{X_j - F, 0\}$, debt $\min\{X_j, F\}$.
- Agency cost: expected firm payoff of the value-maximizing strategy minus that of the strategy equity chooses.

**Assumption.** Risk is diversifiable, so every claim is discounted at the riskless rate, which is not given. All comparisons are in expected payoffs at the payoff date; present values are these divided by the same factor $1 + r_f$, so the rankings do not change.

**Step 1 — (a) Expected firm payoffs.** We weight each payoff by its probability:

$$\mathbb{E}[X_A] = (1.0)(75) = 75, \tag{24}$$
$$\mathbb{E}[X_B] = (0.5)(140) + (0.5)(0) = 70 + 0 = 70, \tag{25}$$
$$\mathbb{E}[X_C] = (0.1)(300) + (0.9)(40) = 30 + 36 = 66. \tag{26}$$

Strategy A has the highest expected payoff (and the lowest risk).

**Answer (a).** Strategy A, with an expected payoff of \$75 million (B: \$70 million; C: \$66 million).

**Step 2 — (b) Debt of \$40 million.** We split each state's payoff with $F = 40$: equity $\max\{X - 40, 0\}$ (e.g. $140 - 40 = 100$; $\max\{40 - 40, 0\} = 0$), debt $\min\{X, 40\}$:

| Strategy, state (prob.) | Firm | Equity $\max\{X-40,0\}$ | Debt $\min\{X,40\}$ |
|---|---|---|---|
| A (1.0) | 75 | 35 | 40 |
| B, high (0.5) | 140 | 100 | 40 |
| B, low (0.5) | 0 | 0 | 0 |
| C, high (0.1) | 300 | 260 | 40 |
| C, low (0.9) | 40 | 0 | 40 |

$$\begin{aligned}\mathbb{E}[E_A] &= (1.0)(35) = 35,\\ \mathbb{E}[E_B] &= (0.5)(100) + (0.5)(0) = 50 + 0 = 50,\\ \mathbb{E}[E_C] &= (0.1)(260) + (0.9)(0) = 26 + 0 = 26.\end{aligned} \tag{27}$$

Equity prefers B. Debt then expects $(0.5)(40) + (0.5)(0) = 20$, against 40 under A or C.

**Answer (b).** Strategy B (expected equity payoff \$50 million, against \$35 million for A and \$26 million for C).

**Step 3 — (c) Debt of \$110 million.** Same split with $F = 110$: equity $\max\{X - 110, 0\}$ (e.g. $\max\{75 - 110, 0\} = 0$, $140 - 110 = 30$, $300 - 110 = 190$), debt $\min\{X, 110\}$:

| Strategy, state (prob.) | Firm | Equity $\max\{X-110,0\}$ | Debt $\min\{X,110\}$ |
|---|---|---|---|
| A (1.0) | 75 | 0 | 75 |
| B, high (0.5) | 140 | 30 | 110 |
| B, low (0.5) | 0 | 0 | 0 |
| C, high (0.1) | 300 | 190 | 110 |
| C, low (0.9) | 40 | 0 | 40 |

$$\begin{aligned}\mathbb{E}[E_A] &= (1.0)(0) = 0,\\ \mathbb{E}[E_B] &= (0.5)(30) + (0.5)(0) = 15 + 0 = 15,\\ \mathbb{E}[E_C] &= (0.1)(190) + (0.9)(0) = 19 + 0 = 19.\end{aligned} \tag{28}$$

Equity prefers C. Debt then expects $(0.1)(110) + (0.9)(40) = 11 + 36 = 47$, against 75 under A and $(0.5)(110) + (0.5)(0) = 55$ under B.

**Answer (c).** Strategy C (expected equity payoff \$19 million, against \$15 million for B and \$0 for A).

**Step 4 — (d) Expected agency costs.** The agency cost is the expected firm payoff of the efficient strategy minus that of the strategy equity picks. The value-maximizing strategy is A (24). With $F = 40$ equity picks B (25); with $F = 110$ it picks C (26):

$$\text{cost}(40) = \mathbb{E}[X_A] - \mathbb{E}[X_B] = (75) - (70) = 5, \tag{29}$$

$$\text{cost}(110) = \mathbb{E}[X_A] - \mathbb{E}[X_C] = (75) - (66) = 9. \tag{30}$$

The expected agency cost is \$5 million with \$40 million of debt and \$9 million with \$110 million (in payoff-date dollars; divide by $1 + r_f$ for today's value).

**Answer (d).** \$5 million with \$40 million of debt; \$9 million with \$110 million of debt (expected payoff-date values).

**Intuition.** With \$40 million due, A leaves equity a safe \$35 million, but B's 50% chance of \$100 million is worth more to equity because the creditors bear the zero-payoff state. With \$110 million due, A leaves equity nothing and only the long shot C pays off above the face value often enough. More debt pushes equity towards riskier, lower-value strategies, and the agency cost grows with leverage. Creditors who anticipate the choice price it in, so the shareholders bear the cost ex ante.

**Common mistakes.**

- Measuring the agency cost as the creditors' loss (e.g. $40 - 20 = 20$ with $F = 40$). That is a transfer; the agency cost is the loss in *total* expected payoff, $75 - 70 = 5$.
- Giving equity $X - F$ in states where $X < F$ (C's low state with $F = 110$ gives $-70$, not 0), which makes the risky strategies look worse to equity than they are.
- Comparing equity's payoffs across debt levels instead of across strategies at a given debt level; each of (b) and (c) is a choice among A, B, C with $F$ fixed.

**Ref:** Berk Ch. 16, § 16.6 "Agency Costs and Information", "Equity-Debt Holder Conflicts" (excessive risk-taking), pp. 483–484. Lecture Topic 4B, slides 2–13.

## 7. Ajax Manufacturing: debt overhang — Grinblatt & Titman 16.12–16.13, p. 593

**What is asked.** (16.12) The NPV of the widget project, and whether Ajax can raise the \$75 million when the existing \$70 million senior bond forbids new debt (so only new equity is possible). (16.13) Whether Ajax can raise the \$75 million in equity if it uses the riskier, capital-intensive process instead.

**Approach.** Risk neutrality and $r_f = 0$ make every value an expected payoff. New equity is junior to the bond, so the most new shareholders can be offered is the whole residual claim $\mathbb{E}[\max\{X - 70, 0\}]$; the project is fundable iff that is at least \$75 million. This is the lecture's *debt overhang*: the bond captures part of the project's value. In 16.13 a riskier payoff shrinks the bond's share (*asset substitution*), which can make the equity issue feasible.

**Notation.**

- $F = 70$: existing senior bond obligation, with a covenant forbidding additional debt; $I = 75$: investment cost.
- States: recession (probability $\pi = 0.2$) and favourable (probability 0.8); $X_s$: project cash flow in state $s$; risk-neutral investors, $r_f = 0$, no taxes or bankruptcy costs.
- $V$: project value; $E^{\max}$: value of the whole residual claim after the bond, the most new shareholders can be offered; $D$: bond value.

**Assumption.** The firm has no other assets: without the project, the bond receives nothing. In 16.13 the investment still costs $I = 75$ (the statement changes only the cash flows).

**Step 1 — (16.12) NPV of the project.** We value the project's cash flows and subtract their cost:

$$\begin{aligned} V &= \pi X_{\text{rec}} + (1-\pi)X_{\text{fav}} \qquad (31)\\ &= (0.2)(85) + (0.8)(150)\\ &= 17 + 120 = 137,\\ NPV &= V - I = (137) - (75) = 62. \qquad (32)\end{aligned}$$

**Step 2 — (16.12) Can it be funded?.** We value the largest claim new shareholders can buy and compare it with the \$75 million they must pay. New debt is forbidden, so the money must come from equity, which is junior to the \$70 bond. Equity receives $\max\{85-70,0\} = 15$ in a recession and $\max\{150-70,0\} = 80$ in the favourable state:

$$\begin{aligned} E^{\max} &= (0.2)(15) + (0.8)(80) \qquad (33)\\ &= 3 + 64 = 67 < 75 = I.\end{aligned}$$

Even 100% of the equity is worth less than the \$75 million it must raise: **no**. The bond, which would be paid in full ($85 \geq 70$), would be worth $D = 70$ instead of 0; the split $70 + (67 - 75) = 62$ is the NPV. The creditors capture more than the whole NPV, which is why the shareholders cannot pay for it.

**Answer (16.12).** $NPV = \$62$ million, but the firm **cannot** fund the project: all of the equity is worth \$67 million $<$ \$75 million.

**Step 3 — (16.13) The capital-intensive variant.** We repeat Steps 1–2 with the new cash flows, 5 (recession) and 170 (favourable):

$$\begin{aligned} V' &= (0.2)(5) + (0.8)(170)\\ &= 1 + 136 = 137,\\ NPV' &= (137) - (75) = 62,\end{aligned}$$

the same NPV but riskier. Equity receives $\max\{5-70,0\} = 0$ and $170 - 70 = 100$:

$$\begin{aligned} E'^{\max} &= (0.2)(0) + (0.8)(100) \qquad (34)\\ &= 80 \geq 75 = I.\end{aligned}$$

**Yes**: new shareholders pay \$75 million for a fraction $75/80 = 0.9375$ of the equity and the old shareholders keep $1 - 0.9375 = 6.25\%$, worth $0.0625(80) = 80 - 75 = 5$. The bond now receives $\min\{5, 70\} = 5$ and $\min\{170, 70\} = 70$, worth

$$\begin{aligned} D' &= (0.2)(5) + (0.8)(70)\\ &= 1 + 56 = 57,\end{aligned}$$

and $57 + 80 = 137 = V'$.

**Answer (16.13).** **Yes**: under the capital-intensive process the equity is worth \$80 million $\geq$ \$75 million, so new shareholders buy 93.75% of it for \$75 million.

**Step 4 — Comparison.** The two processes create the same value, 62. The risky one moves $70 - 57 = 13$ from the bondholders to the equity, which is exactly what lifts the equity's value from 67 to 80, above the \$75 million cost. The bondholders prefer the safe process, but under it nothing is built and they get 0, so they are better off with the risky one ($57 > 0$).

**Intuition.** Debt overhang: with a senior bond that cannot be diluted, new equity must pay for the whole project but receives only the value left after the bond, and here the bond absorbs too much of it. Raising risk shrinks the bond's share (asset substitution), and that makes the equity issue possible.

Of the two distortions, the risk shift is the lesser here. A cleaner fix is to renegotiate the bond: under the safe process equity covers \$75 million whenever the face is cut to at most $137 - 75 = 62$, which the bondholders should accept since $62 > 57$ (the lecture's renegotiation argument).

**Common mistakes.**

- Answering "NPV $> 0$, so yes" in 16.12. Fundability depends on what the junior claim is worth to the new money, not on total value.
- Valuing equity as $V - F$. That gives $137 - 70 = 67$ for *both* processes and the wrong answer in 16.13; limited liability makes the recession payoff $\max\{5 - 70, 0\} = 0$, not $-65$.
- Forgetting that without the project the bond gets nothing here; that is why the bondholders prefer the risky process (57) to no project (0).

**Ref:** Grinblatt & Titman Ch. 16, § 16.2 "The Debt Overhang Problem" and "How the Seniority Structure of Debt Affects the Debt Overhang Problem", pp. 563–566 (Results 16.3–16.4); "The Asset Substitution Problem", pp. 569–572. Lecture Topic 4B, slides 19–31.

## 8. Project riskiness and credit rationing — Tirole 3.15, p. 150

**What is asked.** Two versions of the fixed-investment project have the same cost, the same expected income under good behaviour and the same effect of shirking on the success probability, but B succeeds less often with a larger payoff. Decide which version can be financed for a wider range of the entrepreneur's cash $A$, i.e. is less prone to credit rationing.

**Approach.** In the Holmström–Tirole fixed-investment model (lecture Topic 4A) a project is financed iff $A \geq \bar{A} = I - \mathcal{P}$, where $\mathcal{P} = p_H(R - B/\Delta p)$ is the *pledgeable income*. So the variant with the lower threshold $\bar{A}$, i.e. the higher pledgeable income, is less rationed. We write $\bar{A}$ as non-pledgeable rent minus NPV, use equal NPVs, and compare the rents.

**Notation.**

- $j \in \{A, B\}$: project variant; $p_H^j > p_L^j$: success probabilities with good / bad behaviour; $R^j > 0$: income on success (0 on failure).
- $\Delta p = p_H^A - p_L^A = p_H^B - p_L^B > 0$: common; $B > 0$: private benefit (common); $I$: common investment; $A$: borrower's cash.
- Maintained: $p_H^A R^A = p_H^B R^B$ (equal NPV) and $p_H^A > p_H^B$ (B riskier); (3.1), (3.2) hold for both variants.
- $\mathcal{P}^j = p_H^j(R^j - B/\Delta p)$: pledgeable income; $\bar{A}^j$: minimum cash for financing variant $j$ (3.3).

**Step 1 — The minimum stake is the same.** We first derive the borrower's minimum reward in each variant. Variant $j$'s incentive constraint is $p_H^j R_b^j \geq p_L^j R_b^j + B$, i.e. $(p_H^j - p_L^j)R_b^j \geq B$, i.e. $R_b^j \geq B/\Delta p$: the same for both variants, because $B$ and $\Delta p$ are common.

**Step 2 — Financing thresholds.** We compute each variant's threshold from $(\mathrm{IR}_l)$, $\mathcal{P}^j \geq I - A$. From (3.3),

$$\begin{aligned} \bar{A}^j &= I - \mathcal{P}^j \\ &= I - p_H^j \left( R^j - \frac{B}{\Delta p} \right) \\ &= I - p_H^j R^j + p_H^j \frac{B}{\Delta p} \\ &= p_H^j \frac{B}{\Delta p} - (p_H^j R^j - I). \end{aligned} \tag{35}$$

The first term is the non-pledgeable income: the expected rent the borrower must be left.

**Step 3 — Compare.** Subtracting (35) for B from that for A, and substituting $p_H^B R^B = p_H^A R^A$,

$$\begin{aligned}\bar{A}^A - \bar{A}^B &= \left[p_H^A \frac{B}{\Delta p} - (p_H^A R^A - I)\right] - \left[p_H^B \frac{B}{\Delta p} - ((p_H^A R^A) - I)\right] \\ &= p_H^A \frac{B}{\Delta p} - p_H^A R^A + I - p_H^B \frac{B}{\Delta p} + p_H^A R^A - I \\ &= (p_H^A - p_H^B)\frac{B}{\Delta p} > 0, \end{aligned} \tag{36}$$

positive because $p_H^A > p_H^B$, $B > 0$, $\Delta p > 0$. The riskier variant B is financed for every $A \geq \bar{A}^B$, including $A \in [\bar{A}^B, \bar{A}^A)$ where A is not: **variant B is less prone to credit rationing**.

**Answer (3.15).** Variant B, the riskier one: $\bar{A}^B = \bar{A}^A - (p_H^A - p_H^B)B/\Delta p < \bar{A}^A$, so every entrepreneur financed under A is financed under B, and those with $A \in [\bar{A}^B, \bar{A}^A)$ only under B (given that both variants need good behaviour, (3.2)).

**Step 4 — Numerical illustration.** $I = 100$, $\Delta p = 0.2$, $B = 9$; A: $p_H^A = 0.8$, $p_L^A = 0.6$, $R^A = 150$; B: $p_H^B = 0.6$, $p_L^B = 0.4$, $R^B = 200$. Both NPVs are $120 - 100 = 20$; (3.2) holds ($0.6(150) - 100 + 9 = -1 < 0$ and $0.4(200) - 100 + 9 = -11 < 0$); $B/\Delta p = 45$. Then $\mathcal{P}^A = 0.8(105) = 84$, $\bar{A}^A = 16$; $\mathcal{P}^B = 0.6(155) = 93$, $\bar{A}^B = 7$; and $\bar{A}^A - \bar{A}^B = (0.2)(45) = 9$, as (36) says.

**Step 5 — Beyond the maintained assumption (the book's extension).** If (3.2) fails for the safe variant, a contract that leaves all income to investors ($R_b = 0$, the borrower misbehaves) may finance it: investors then receive $p_L^A R^A$, and the borrower, who gets the NPV with shirking, $p_L^A R^A + B - I$, when investors break even, wants it if that NPV is positive. The safe variant can then be financed and the risky one not when (in order) B cannot be financed with good behaviour, B cannot be financed with shirking, A can be financed with shirking, and A's NPV with shirking is positive:

$$I - A > p_H^B \left(R^B - \frac{B}{\Delta p}\right), \quad I - A > p_L^B R^B, \quad I - A \leq p_L^A R^A, \quad I < p_L^A R^A + B. \tag{37}$$

These conditions are compatible. With $A = 0$, $I = 120$, $\Delta p = 0.3$, $B = 45$, $p_H^A = 0.9$, $R^A = 200$, $p_H^B = 0.5$, $R^B = 360$ (both $p_H R = 180$): $p_H^B(R^B - B/\Delta p) = 0.5(210) = 105 < 120$; $p_L^B R^B = 0.2(360) = 72 < 120$; $p_L^A R^A = 0.6(200) = 120 \geq 120$; $p_L^A R^A + B = 165 > 120$. Here the risky variant is rationed and the safe one is funded. Good behaviour adds more pledgeable income when the success payoff is large, which favours the risky variant only when good behaviour is needed for funding.

**Intuition.** The borrower's rent $B/\Delta p$ is paid only on success. A safer project succeeds more often, so it pays out the rent more often and leaves less expected income to pledge: the non-pledgeable rent $p_H B/\Delta p$ is higher for the same NPV. Riskier projects with the same mean concentrate income in success, where it is pledged anyway.

**Common mistakes.**

- Importing the Stiglitz–Weiss intuition that riskier borrowers are rationed first. Here riskiness lowers the rent $p_H B/\Delta p$ and so *raises* pledgeable income.
- Computing the threshold as $I - p_H R$ (all income pledged), which is the same for both variants by $p_H^A R^A = p_H^B R^B$ and hides the difference.
- Forgetting that the comparison holds only while good behaviour is required for funding: if (3.2) fails for one variant, the book's extension (Step 5) can reverse the ranking.

**Book answer.** Matches the book's answer (p. 578): with the same minimum reward $B/\Delta p$, the safe project A is financed for a smaller range of $A$, i.e. it is more prone to credit rationing; the book then shows, with conditions (37), that the risky project may be the more rationed one if high-powered incentives are not necessarily called for. One slip in the book's prose: it says the third condition

means that "the risky project" generates enough pledgeable income when investors get all the cash flow, but that condition, $I - A \le p_L^A R^A$, is about the safe variant A.

**Ref:** Tirole Ch. 3, § 3.2.1–3.2.2 "The Fixed-Investment Model", "The Lenders' Credit Analysis", pp. 115–117. Lecture Topic 4A, slides 26–35 (determinants of credit rationing).

## 9. Entrepreneurial risk aversion — Tirole Exercise 3.2, p. 144

**What is asked.** (i) The entrepreneur is risk neutral above a subsistence level $c_0$ and infinitely averse to falling below it. Find the smallest cash $\bar{A}$ with which risk-neutral investors finance the project, and explain why the answer differs between a project that succeeds for sure under effort ($p_H = 1$) and one that can fail even under effort ($p_H < 1$). (ii) Redo the analysis for a general strictly concave utility, once with limited liability (her payments cannot be negative) and once without.

**Approach.** This is the fixed-investment model of the lecture (Topic 4A) with a risk-averse borrower. The contract pays her $c^S$ on success and $c^F$ on failure; she chooses it to maximise expected utility subject to the investors' breakeven constraint ($\text{IR}_l$) and her incentive constraint ($\text{IC}_b$), now written in utility. Pledgeable income is total expected income minus the cheapest expected payment that satisfies ($\text{IC}_b$) and the lower bound on $c^F$; a floor on consumption (subsistence in (i), limited liability in (ii)) raises that payment and so the minimum equity. In (ii) we write the Lagrangian and derive the first-order conditions, which give the Holmström–Shavell insurance-versus-incentives trade-off.

**Notation.**

- $A$: entrepreneur's cash ($A \ge c_0$ in (i)); $I > A$: fixed investment; $R > 0$: income in the case of success, 0 in the case of failure.
- $p_H$, $p_L = p_H - \Delta p$: success probabilities when she works / shirks ($0 < p_L < p_H \le 1$); $B > 0$: private benefit from shirking.
- $NPV \equiv p_H R - I > 0$; $\bar{A}_{rn} \equiv I - p_H(R - B/\Delta p) > 0$: the risk-neutral threshold (3.3), positive by the assumption $I > p_H(R - B/\Delta p)$.
- $c^S$, $c^F$: the entrepreneur's consumption in the case of success / failure (her total payoff from the contract); $u(\cdot)$: utility of consumption; $c_0$: subsistence level in (i).
- $\mathcal{P}$: expected pledgeable income; $\bar{A}$: minimum cash for financing.
- $h(c) \equiv u^{-1}(u(c) + B/\Delta p)$: consumption in success that gives exactly the utility premium $B/\Delta p$ over consumption $c$ in failure (part (ii)).

**Assumption.** The entrepreneur puts all of $A$ into the project and the contract pays her $c^S$ or $c^F$ (keeping cash aside is the same as adding it to both payments). In (ii) the private benefit enters utility additively, $p\,u(c^S) + (1-p)\,u(c^F) + B$ when she shirks, as in the book's answer.

**Step 1 — (i), $p_H < 1$: the programme.** The entrepreneur offers the contract that maximises her expected utility subject to the investors' breakeven and her incentive constraint:

$$\max_{c^S, c^F} \; p_H\,u(c^S) + (1 - p_H)\,u(c^F) \tag{38}$$

$$\text{s.t.} \quad p_H(R - c^S) + (1 - p_H)(0 - c^F) \;\ge\; I - A, \tag{39}$$

$$p_H\,u(c^S) + (1 - p_H)\,u(c^F) \;\ge\; p_L\,u(c^S) + (1 - p_L)\,u(c^F) + B. \tag{40}$$

Failure has probability $1 - p_H > 0$ even under work, so $c^F < c_0$ would make (38) equal to $-\infty$: the optimum has $c^F \ge c_0$, and then $c^S \ge c_0$ too. On $[c_0, \infty)$ utility is linear, $u(c) = c$, so (40) reads

$$\begin{aligned} p_H c^S + (1 - p_H)c^F &\ge p_L c^S + (1 - p_L)c^F + B \\ (p_H - p_L)c^S - [(1 - p_L) - (1 - p_H)]c^F &\ge B \\ (p_H - p_L)c^S - (p_H - p_L)c^F &\ge B \\ (\Delta p)(c^S - c^F) &\ge B. \end{aligned} \tag{41}$$

**Step 2 — (i), $p_H < 1$: which constraints bind, and $\bar{A}$.** With linear utility the objective (38) is the expected payment $p_H c^S + (1-p_H)c^F$. If (39) were slack, raising $c^S$ would raise the objective and relax (41); so (39) binds and the entrepreneur gets $A + NPV$. Financing is possible if and only if the cheapest contract that satisfies (41) and $c^F \geq c_0$ leaves the investors enough. Using $c^S \geq c^F + B/\Delta p$ from (41) and then $c^F \geq c_0$, any feasible contract costs

$$\begin{aligned} p_H c^S + (1-p_H)c^F &\geq p_H\Big(c^F + \frac{B}{\Delta p}\Big) + (1-p_H)c^F \\ &= c^F + p_H \frac{B}{\Delta p} \;\geq\; c_0 + p_H \frac{B}{\Delta p}, \end{aligned}$$

with equality iff both inequalities bind, so the cheapest contract is

$$c^F = c_0, \tag{42}$$

$$c^S = c_0 + \frac{B}{\Delta p}. \tag{43}$$

The pledgeable income is total expected income minus this minimum expected payment; substituting (42)–(43),

$$\begin{aligned} \mathcal{P} &= p_H R - \Big[p_H\Big(c_0 + \frac{B}{\Delta p}\Big) + (1-p_H)(c_0)\Big] \\ &= p_H\Big(R - \frac{B}{\Delta p}\Big) - c_0. \end{aligned} \tag{44}$$

(the middle step: $p_H c_0 + (1-p_H)c_0 = c_0$). Financing requires $\mathcal{P} \geq I - A$; using (44) and moving terms,

$$\begin{aligned} p_H\Big(R - \frac{B}{\Delta p}\Big) - c_0 &\geq I - A \\ A \;\geq\; \bar{A} &= I + c_0 - p_H\Big(R - \frac{B}{\Delta p}\Big) \\ &= \Big[I - p_H\Big(R - \frac{B}{\Delta p}\Big)\Big] + c_0 = \bar{A}_{rn} + c_0. \end{aligned} \tag{45}$$

Because $\bar{A}_{rn} > 0$, $\bar{A} > c_0$: entrepreneurs with $A \in [c_0, \bar{A})$ are rationed. The participation constraint holds, since she gets $A + NPV > A$. Illustration: with $p_H = 0.8$, $p_L = 0.4$, $B = 1$, $R = 10$, $I = 6.5$, $c_0 = 1$, (45) gives $\bar{A} = (6.5) + (1) - (0.8)(10 - 1/0.4) = 1.5$, against $\bar{A}_{rn} = 0.5$.

**Step 3 — (i), $p_H = 1$.** Now failure never happens on the equilibrium path; it happens only if she shirks, with probability $1 - p_L > 0$. Set $c^F < c_0$ (for example $c^F = 0$, feasible under limited liability when $c_0 > 0$). Shirking then yields $p_L\, u(c^S) + (1-p_L)(-\infty) + B = -\infty$, so (40) holds for every $c^S \geq c_0$ and the incentive problem costs nothing. The cheapest contract is $c^S = c_0$, and

$$\mathcal{P} = R - c_0, \tag{46}$$

$$\bar{A} = I + c_0 - R \;<\; c_0, \tag{47}$$

where the inequality uses $R - I = NPV > 0$. Every entrepreneur with $A \geq c_0$ is financed: no rationing (with the numbers above, $\bar{A} = 6.5 + 1 - 10 = -2.5$). The threshold is discontinuous at $p_H = 1$: by (45), $\bar{A} \to I + c_0 - R + B/\Delta p$ as $p_H \to 1$, a jump of $B/\Delta p$. If $c_0 \leq 0$, limited liability rules out a punishment below $c_0$ and the risk-neutral analysis applies.

**Answer (i).** For $p_H < 1$: $\bar{A} = I + c_0 - p_H(R - B/\Delta p) = \bar{A}_{rn} + c_0$, so entrepreneurs with $A \in [c_0, \bar{A})$ are rationed. For $p_H = 1$: failure reveals shirking and can be punished with $c^F < c_0$ at no cost on the equilibrium path, so $\bar{A} = I + c_0 - R < c_0$ and every entrepreneur with $A \geq c_0$ is financed (no rationing).

**Step 4 — (ii) The general programme.** With $u' > 0$, $u'' < 0$, the contract solves

$$\max_{c^S, c^F} \; p_H\, u(c^S) + (1 - p_H)\, u(c^F) \tag{48}$$

$$\text{s.t.} \quad p_H(R - c^S) - (1 - p_H)c^F \;\geq\; I - A, \tag{IR$_l$}$$

$$(\Delta p)\big[u(c^S) - u(c^F)\big] \;\geq\; B, \tag{IC$_b$}$$

$$c^F \geq 0,\; c^S \geq 0 \quad \text{(only under limited liability)}, \tag{LL}$$

$$p_H\, u(c^S) + (1 - p_H)\, u(c^F) \;\geq\; u(A). \tag{PC}$$

(IC$_b$) is the risk-averse version of (IC$_b$): subtract $p_L u(c^S) + (1 - p_L)u(c^F) + B$ from (48). (PC) says she prefers the project to consuming $A$. In utility units $v^S = u(c^S)$, $v^F = u(c^F)$ the objective and (IC$_b$) are linear and the cost $p_H u^{-1}(v^S) + (1 - p_H)u^{-1}(v^F)$ is convex, so the problem is convex and the first-order conditions are sufficient.

**Step 5 — (ii) Which constraints bind.** We show both (IR$_l$) and (IC$_b$) bind, the second by deriving the first-order conditions. (a) (IR$_l$) binds: otherwise raise $c^S$, which raises (48) and relaxes (IC$_b$). (b) With multipliers $\lambda \geq 0$ on (IR$_l$) and $\mu \geq 0$ on (IC$_b$) (and (LL) slack), the Lagrangian is

$$\begin{aligned}\mathcal{L} = {} & p_H\, u(c^S) + (1 - p_H)\, u(c^F) + \lambda\big[p_H(R - c^S) - (1 - p_H)c^F - I + A\big] \\ & + \mu\big[(\Delta p)\big(u(c^S) - u(c^F)\big) - B\big].\end{aligned}$$

Differentiating with respect to $c^S$ and $c^F$ and setting the derivatives to zero,

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial c^S} &= p_H u'(c^S) - \lambda p_H + \mu(\Delta p)u'(c^S) = 0, \\ &\Longrightarrow \quad u'(c^S)\big(p_H + \mu\Delta p\big) = \lambda p_H, \\ \frac{\partial \mathcal{L}}{\partial c^F} &= (1 - p_H)u'(c^F) - \lambda(1 - p_H) - \mu(\Delta p)u'(c^F) = 0, \\ &\Longrightarrow \quad u'(c^F)\big(1 - p_H - \mu\Delta p\big) = \lambda(1 - p_H).\end{aligned}$$

Here $\lambda > 0$, because $\lambda = 0$ would make the first condition read $u'(c^S)(p_H + \mu\Delta p) = 0$, impossible with $u' > 0$. Dividing the first by $\lambda p_H\, u'(c^S)$ and the second by $\lambda(1 - p_H)\, u'(c^F)$,

$$\frac{1}{u'(c^S)} = \frac{1}{\lambda}\Big(1 + \mu\frac{\Delta p}{p_H}\Big), \tag{49}$$

$$\frac{1}{u'(c^F)} = \frac{1}{\lambda}\Big(1 - \mu\frac{\Delta p}{1 - p_H}\Big). \tag{50}$$

If $\mu = 0$, (49)–(50) give $u'(c^S) = u'(c^F)$, i.e. full insurance $c^S = c^F$, which violates (IC$_b$) because $B > 0$. So $\mu > 0$ and, by (49)–(50), $c^S > c^F$: the entrepreneur bears risk, but only as much as incentives require (Holmström–Shavell inverse-marginal-utility rule).

**Step 6 — (ii) The solution.** We reduce the two binding constraints to one equation in $c^F$. The binding (IC$_b$) is $u(c^S) = u(c^F) + B/\Delta p$, i.e. $c^S = h(c^F)$. The binding (IR$_l$) is $p_H R - p_H c^S - (1 - p_H)c^F = I - A$, i.e.

$$p_H c^S + (1 - p_H)c^F = A + (p_H R - I) = A + NPV.$$

Substituting $c^S = h(c^F)$,

$$p_H\, h(c^F) + (1 - p_H)\, c^F = A + NPV. \tag{51}$$

Differentiating the identity $u(h(c)) = u(c) + B/\Delta p$ with respect to $c$,

$$u'\big(h(c)\big)\, h'(c) = u'(c) \quad \Longrightarrow \quad h'(c) = \frac{u'(c)}{u'(h(c))} \;\geq\; 1,$$

because $h(c) > c$ and $u'$ is decreasing. Hence the left-hand side of (51) is increasing in $c^F$, so (51) defines a unique $c^F = c(A)$, increasing in $A$. Substituting $u(c^S) = u(c^F) + B/\Delta p$ into (48), her utility is

$$\begin{aligned} V(A) &= p_H\Big(u(c(A)) + \frac{B}{\Delta p}\Big) + (1 - p_H)\, u(c(A)) \\ &= u(c(A)) + \frac{p_H B}{\Delta p}. \end{aligned} \tag{52}$$

**Step 7 — (ii) Without limited liability.** Only (PC) can fail. Let $CE$ be the certainty equivalent of the lottery $(c^S, c^F)$, $u(CE) = V(A)$, and $\pi \equiv \mathbb{E}[c] - CE$ its risk premium; by (51), $\mathbb{E}[c] = A + NPV$. Then

$$\text{financing} \iff V(A) \geq u(A) \iff CE \geq A \iff \pi(A) \leq NPV. \tag{53}$$

A risk-neutral entrepreneur ($\pi = 0$) without limited liability is always financed (she can be punished with $c^F < 0$; there is no $\bar{A}$). A risk-averse one is financed only if the risk premium of the incentive lottery does not exceed the NPV: risk aversion alone can block positive-NPV projects. Whether this gives a threshold in $A$ depends on how $\pi$ varies with wealth (with $B$ in utility units, a richer entrepreneur needs a larger money spread to produce the utility gap $B/\Delta p$).

**Step 8 — (ii) With limited liability.** (LL) requires $c(A) \geq 0$. Because the left-hand side of (51) is increasing, $c(A) \geq 0$ iff $A + NPV \geq p_H h(0)$, i.e.

$$A \;\geq\; A_{LL} \equiv I - p_H\big[R - h(0)\big]. \tag{54}$$

Below $A_{LL}$ no contract satisfies ($\text{IR}_l$), ($\text{IC}_b$) and (LL): the cheapest incentive-compatible contract, $c^F = 0$, $c^S = h(0)$, already costs more than $A + NPV$. Above $A_{LL}$ the solution is that of Step 6 with $c^F = c(A) \geq 0$, so the minimum equity is $\bar{A} = \min\{A \geq A_{LL} :\ V(A) \geq u(A)\}$. Two remarks. First, for $A > A_{LL}$ the entrepreneur receives $c^F > 0$ in failure: unlike the risk-neutral case, pure debt ($c^F = 0$) is no longer optimal, because she buys partial insurance. Second, risk neutrality gives back the book: with $u(c) = c$, $h(0) = B/\Delta p$ and (54) is (45) with $c_0 = 0$; part (i) is the case of a floor $c_0$ instead of 0. If $u$ is normalised so that $u'(0) = 1$, concavity gives $h(0) \geq B/\Delta p$, so $A_{LL} \geq \bar{A}_{rn}$: risk aversion raises the equity requirement.

**Answer (ii).** Both ($\text{IR}_l$) and ($\text{IC}_b$) bind; the contract solves (51) with $u(c^S) = u(c^F) + B/\Delta p$, and the first-order conditions (49)–(50) give $c^S > c^F$ (partial insurance). Without limited liability, financing obtains iff the risk premium of the incentive lottery is at most the NPV, $\pi(A) \leq NPV$. With limited liability, also $A \geq A_{LL} = I - p_H[R - h(0)]$, and $\bar{A} = \min\{A \geq A_{LL} : V(A) \geq u(A)\}$; above $A_{LL}$ she gets $c^F > 0$, so pure debt is no longer optimal.

**Intuition.** Incentives need a spread between the success and failure payoffs. A risk-neutral entrepreneur bears that spread for free; a subsistence floor or limited liability bounds the punishment, so the spread must be paid as a rent in success, and a risk-averse entrepreneur must in addition be compensated for the risk. Both raise the stake she keeps and lower pledgeable income. The $p_H = 1$ case is the exception: failure then proves shirking, so an unbounded punishment off the equilibrium path removes the agency cost.

**Common mistakes.**

- Subtracting $c_0$ from the cash instead of adding it to the payments: the entrepreneur must be paid at least $c_0$ in *both* states, which costs investors $c_0$ in expectation; the two bookkeepings agree only if this is done consistently.
- Treating $p_H = 1$ like $p_H < 1$. With $p_H = 1$ failure never occurs under effort, so the $-\infty$ punishment is free; with $p_H < 1$ it would hit her on the equilibrium path.
- In (ii), carrying over the risk-neutral result that pure debt ($c^F = 0$) is optimal. A risk-averse borrower buys partial insurance, $c^F > 0$, whenever $A > A_{LL}$.

**Book answer.** Matches the book's answer (p. 571): $\bar{A} = I + c_0 - p_H(R - B/\Delta p)$ for $p_H < 1$, the programme of (ii) with binding incentive constraint. For $p_H = 1$ the book writes the pledgeable income as $p_H R$, i.e. with $c_0$ kept aside from $A$; this is (46)–(47) in other bookkeeping. The book only states the programme for (ii) and cites Holmström (1979) and Shavell (1979); Steps 6–8 are this solution's.

**Ref:** Tirole §3.2.1–3.2.2, pp. 115–117; answer p. 571.

## 10. Maximal incentives principle in the fixed-investment model — Tirole Exercise 3.18, p. 151

**What is asked.** In the fixed-investment model, failure now yields a positive income $R^F$ (salvage value) instead of 0. Show that a debt contract (investors take everything in failure and a fixed amount in success) is an optimal financial structure, and show that, unlike the variable-investment model of Section 3.4.3, other contracts may be optimal too.

**Approach.** This is the lecture's "debt dominates equity" result (maximal incentives principle, Topic 4A). Set up the borrower's programme: maximise her expected reward subject to ($\text{IC}_b$), ($\text{IR}_l$) and limited liability. With a fixed investment her utility equals the NPV for *every* contract on which ($\text{IR}_l$) binds, so the contract matters only for feasibility, i.e. for *pledgeable income*. Paying her in failure buys no incentive, so the contract that maximises pledgeable income pays her nothing in failure: debt. Uniqueness fails because, when $A > \bar{A}$, the spare pledgeable income can be paid in either state.

**Notation.**

- $I$: fixed investment; $A$: entrepreneur's cash; $R^S > R^F > 0$: income in success / failure, with $R^F < I - A$; $R \equiv R^S - R^F$.
- $p_H$, $p_L = p_H - \Delta p$, $B$: as in Section 3.2.
- $R_b^S$, $R_b^F \geq 0$: entrepreneur's reward in success / failure (limited liability).
- $NPV \equiv p_H R^S + (1 - p_H)R^F - I = p_H R + R^F - I > 0$; $U_b$: entrepreneur's net utility.
- $D$: face value of debt; $\bar{A}$: minimum cash for financing.

**Step 1 — Programme.** The borrower designs the contract $(R_b^S, R_b^F)$; her incentive constraint is $p_H R_b^S + (1 - p_H)R_b^F \geq p_L R_b^S + (1 - p_L)R_b^F + B$, which simplifies (collect $R_b^S$ and $R_b^F$ terms, $(1 - p_L) - (1 - p_H) = \Delta p$) to ($\text{IC}_b$):

$$
\begin{aligned}
U_b = \max_{R_b^S, R_b^F} \; & p_H R_b^S + (1 - p_H)R_b^F - A && (55)\\
\text{s.t.} \quad & (\Delta p)(R_b^S - R_b^F) \;\geq\; B, && (\text{IC}_b)\\
& p_H(R^S - R_b^S) + (1 - p_H)(R^F - R_b^F) \;\geq\; I - A, && (\text{IR}_l)\\
& R_b^S \geq 0, \;\; R_b^F \geq 0. && (\text{LL})
\end{aligned}
$$

**Step 2 — The breakeven constraint binds; $U_b$ is the NPV.** If ($\text{IR}_l$) were slack, raising $R_b^S$ would raise (55) and relax ($\text{IC}_b$). With ($\text{IR}_l$) binding, solve it for her expected reward,

$$
\begin{aligned}
p_H R^S - p_H R_b^S + (1 - p_H)R^F - (1 - p_H)R_b^F &= I - A\\
p_H R_b^S + (1 - p_H)R_b^F &= p_H R^S + (1 - p_H)R^F - (I - A),
\end{aligned}
$$

and substitute it into (55):

$$
\begin{aligned}
U_b &= [p_H R^S + (1 - p_H)R^F - (I - A)] - A\\
&= p_H R^S + (1 - p_H)R^F - I\\
&= NPV. \qquad (56)
\end{aligned}
$$

Because $I$ is fixed, (56) is the same for *every* feasible contract with binding ($\text{IR}_l$). The financial structure matters only for feasibility: it must deliver ($\text{IC}_b$) while leaving investors $I - A$.

**Step 3 — Maximal pledgeable income and the financing condition.** The constraint set is nonempty iff the largest expected income that can go to investors subject to ($\text{IC}_b$) and (LL) covers $I - A$, i.e. iff expected income minus the *smallest* expected reward compatible with ($\text{IC}_b$) and (LL) is at least $I - A$. Using $R_b^S \geq R_b^F + B/\Delta p$ from ($\text{IC}_b$),

$$\begin{aligned} p_H R_b^S + (1-p_H)R_b^F &\geq p_H\Big(R_b^F + \frac{B}{\Delta p}\Big) + (1-p_H)R_b^F \\ &= R_b^F + p_H\frac{B}{\Delta p} \;\geq\; p_H\frac{B}{\Delta p}, \end{aligned}$$

the last step by $R_b^F \geq 0$; both bounds are attained by $R_b^F = 0$, $R_b^S = B/\Delta p$. Each unit of failure reward costs one unit of expected income and buys no incentive. So

$$\begin{aligned} \mathcal{P} &= p_H R^S + (1-p_H)R^F - p_H\frac{B}{\Delta p} \\ &= p_H(R^S - R^F) + R^F - p_H\frac{B}{\Delta p} \\ &= R^F + p_H\Big(R - \frac{B}{\Delta p}\Big), \end{aligned} \tag{57}$$

and financing obtains iff $\mathcal{P} \geq I - A$, i.e.

$$A \;\geq\; \bar{A} \equiv I - R^F - p_H\Big(R - \frac{B}{\Delta p}\Big) = p_H\frac{B}{\Delta p} - NPV, \tag{58}$$

where the second form uses $NPV = p_H R + R^F - I$. The safe income $R^F$ works exactly like cash on hand.

**Step 4 — A debt contract is optimal.** Let investors hold debt of face value $D$ defined by

$$p_H D + (1-p_H)R^F = I - A. \tag{59}$$

In failure the firm defaults and investors take $R^F$; in success they get $D$. By $R^F < I - A$, (59) gives $D > R^F$, so the borrower gets $R_b^F = 0$. Substituting (59) into (58) (written as $p_H(R^S - B/\Delta p) + (1-p_H)R^F \geq I - A$),

$$\begin{aligned} p_H\Big(R^S - \frac{B}{\Delta p}\Big) + (1-p_H)R^F &\geq p_H D + (1-p_H)R^F \\ p_H\Big(R^S - \frac{B}{\Delta p}\Big) &\geq p_H D \\ R^S - \frac{B}{\Delta p} &\geq D \\ R_b^S = R^S - D &\geq \frac{B}{\Delta p}, \end{aligned} \tag{60}$$

(subtracting $(1-p_H)R^F$, then dividing by $p_H > 0$, then rearranging) so ($\text{IC}_b$) holds, ($\text{IR}_l$) binds by construction, and by (56) debt attains the maximum. It is optimal whenever financing is feasible.

**Step 5 — Debt is not uniquely optimal.** The optimal set is every $(R_b^S, R_b^F) \geq 0$ with binding ($\text{IR}_l$) and ($\text{IC}_b$). From the binding ($\text{IR}_l$), $R_b^S = [A + NPV - (1-p_H)R_b^F]/p_H$. Substituting this into

(IC$_b$),

$$
\begin{aligned}
(\Delta p)\Big(\frac{A+NPV-(1-p_H)R_b^F}{p_H}-R_b^F\Big) &\geq B\\
(\Delta p)\,\frac{A+NPV-(1-p_H)R_b^F-p_HR_b^F}{p_H} &\geq B\\
\frac{A+NPV-R_b^F}{p_H} &\geq \frac{B}{\Delta p}\\
A+NPV-R_b^F &\geq p_H\frac{B}{\Delta p}\\
R_b^F &\leq A+NPV-p_H\frac{B}{\Delta p}=A-\bar{A}, \qquad (61)
\end{aligned}
$$

using (58) in the last line. So every $R_b^F \in [0, A-\bar{A}]$ is optimal (add $R_b^F \leq R^F$ if investors' claim must be nonnegative in each state). The contract is unique only at $A=\bar{A}$. For $A>\bar{A}$, a small failure reward (debt plus a little equity, or even all-equity when its implied $R_b^F$ satisfies (61)) does as well. Illustration: $R^S=12$, $R^F=2$, $p_H=0.8$, $p_L=0.4$, $B=1.6$, $I=9$, $A=3$. Then $B/\Delta p = 1.6/0.4 = 4$, $NPV = 0.8(12)+0.2(2)-9 = 9.6+0.4-9 = 1$, $\bar{A}=0.8(4)-1=2.2$, and by (59) $D=(9-3-0.2(2))/0.8 = 5.6/0.8 = 7$ (with $2<7\leq 12-4=8$). The optimal set is $R_b^F \in [0,\ 3-2.2]=[0, 0.8]$; at $R_b^F=0.8$, $R_b^S=(3+1-0.2(0.8))/0.8 = 3.84/0.8 = 4.8$ and (IC$_b$) binds: $0.4(4.8-0.8)=1.6=B$.

**Step 6 — Contrast with variable investment.** In Section 3.4.3 the entrepreneur's utility $(p_HR+R^F-1)I$ rises with $I$, so she expands $I$ until the financing constraint binds, the analogue of $A=\bar{A}$. Then the slack in (61) is zero, $R_b^F=0$ is forced and debt is the unique optimum; any $R_b^F>0$ would waste incentive power and reduce the borrowing capacity (3.16).

**Answer (3.18).** Debt with face value $D$ given by $p_HD+(1-p_H)R^F = I-A$ (so $R^F<D\leq R^S-B/\Delta p$) is optimal whenever the project can be financed, $A\geq\bar{A}=I-R^F-p_H(R-B/\Delta p)$. It is not uniquely optimal: every contract with $R_b^F\in[0, A-\bar{A}]$ and binding (IR$_l$) gives the same utility $NPV$; the optimum is unique only at $A=\bar{A}$.

**Intuition.** Paying the entrepreneur in failure costs expected income but weakens incentives, so it is the worst way to reward her; debt, which gives her nothing in failure, maximises incentives per unit of pay. With a fixed project, once the project is funded there is no further use for pledgeable income, so any leftover slack can be paid out in either state.

**Common mistakes.**

- Concluding that all-equity is never optimal, as in Section 3.4.3. That needs the investment to expand until the financing constraint binds; with a fixed $I$ and $A>\bar{A}$ it does not bind.
- Leaving $R^F$ out of pledgeable income. The salvage value is not subject to moral hazard and works exactly like cash on hand: $\bar{A}$ falls one for one with $R^F$.
- Writing (IC$_b$) as $(\Delta p)R_b^S\geq B$, forgetting that a failure reward $R_b^F$ weakens incentives: what matters is the spread $R_b^S-R_b^F$.

**Book answer.** Matches the book's answer (p. 579), including condition (1) there, $p_H(R-B/\Delta p)\geq I-(A+R^F)$, the debt implementation with $R^F<D<R^S$ and the non-uniqueness when (IR$_l$) holds with slack at $R_b^F=0$; (61) makes the size of the optimal set explicit.

**Ref:** Tirole §3.2.2, pp. 116–117; §3.4.3, pp. 128–129; answer p. 579.

## 11. Retained-earnings benefit — Tirole Exercise 3.11, p. 148

**What is asked.** (i) Find how much one extra unit of retained earnings (the entrepreneur's date-1 profit kept in the firm) is worth to her at date 2, given that it helps finance a second project whose

private benefit $B^2$ is still random. (ii) Show, by argument or example, that the existence of the second project can get the first project financed even though no date-2 income can be pledged at date 1.

**Approach.** Two applications of the fixed-investment model (lecture Topic 4A, investment and internal funds). At date 2, retained earnings play the role of cash $A$ in (IR$_l$): more of them lets the second project be financed for higher values of $B^2$. Their *shadow value* is the derivative of the entrepreneur's gross date-2 utility. At date 1, that continuation value is lost when project 1 fails, so it works as an extra reward for effort: the date-1 incentive constraint needs less cash reward, and date-1 *pledgeable income* rises.

**Notation.**

- Project $t \in \{1,2\}$: investment $I^t$, income $R^t$ in success, probabilities $p_H^t$, $p_L^t = p_H^t - \Delta p^t$, private benefit $B^t$; $NPV^2 \equiv p_H^2 R^2 - I^2 > 0$.
- $A^1$: entrepreneur's cash at date 1; $R_b^1$: her date-1 reward in success (her retained earnings), 0 in failure.
- $B^2 \in [\underline{B}^2, \bar{B}^2]$: random, c.d.f. $F$, density $f$, revealed at the start of date 2.
- $\hat{B}^2(R)$: largest $B^2$ for which project 2 is financed with retained earnings $R$.
- $W(R)$: entrepreneur's expected gross date-2 utility with retained earnings $R$; $\mu(R) \equiv W'(R)$: shadow value of retained earnings ($v$ in §3.4.2).

**Assumption.** No discounting; the entrepreneur's only date-2 wealth is $R_b^1$ (or 0), and date-2 lenders cannot be pledged date-1 income.

**Step 1 — (i) Date-2 financing given retained earnings $R$.** Project 2 is financed iff its pledgeable income covers the outside funding, (IR$_l$) for project 2:

$$p_H^2\Big(R^2 - \frac{B^2}{\Delta p^2}\Big) \geq I^2 - R. \tag{62}$$

Solving (62) for $B^2$ (expand, move $p_H^2 B^2/\Delta p^2$ to the right and $I^2 - R$ to the left, then multiply by $\Delta p^2/p_H^2 > 0$),

$$\begin{aligned} p_H^2 R^2 - \frac{p_H^2 B^2}{\Delta p^2} &\geq I^2 - R \\ \frac{p_H^2 B^2}{\Delta p^2} &\leq p_H^2 R^2 - I^2 + R = NPV^2 + R \\ B^2 \;\leq\; \hat{B}^2(R) &\equiv \frac{\Delta p^2}{p_H^2}(NPV^2 + R). \end{aligned} \tag{63}$$

By the assumption $\underline{B}^2 > \Delta p^2\, NPV^2/p_H^2$, (63) gives $\hat{B}^2(0) < \underline{B}^2$: with no retained earnings, project 2 is never financed.

**Step 2 — (i) Gross date-2 utility.** We follow the hint and compute her expected gross utility. If financed, project 2's lenders break even and the entrepreneur gets her cash plus $NPV^2$ (equation (3.6)); otherwise she keeps $R$. With probability $F(\hat{B}^2(R))$ it is financed, so

$$\begin{aligned} W(R) &= F\big(\hat{B}^2(R)\big)\big(R + NPV^2\big) + \Big[1 - F\big(\hat{B}^2(R)\big)\Big]R \\ &= R + F\big(\hat{B}^2(R)\big)\, NPV^2. \end{aligned} \tag{64}$$

**Step 3 — (i) Shadow value.** Differentiating (64) and using $d\hat{B}^2/dR = \Delta p^2/p_H^2$ from (63),

$$\begin{aligned} \mu(R) = W'(R) &= 1 + f\big(\hat{B}^2(R)\big)\,\frac{d\hat{B}^2}{dR}\, NPV^2 \\ &= 1 + f\big(\hat{B}^2(R)\big)\,\frac{\Delta p^2}{p_H^2}\, NPV^2 \;\geq 1. \end{aligned} \tag{65}$$

One unit of retained earnings is worth one unit of consumption plus the expected NPV it unlocks: it raises the probability of financing at the margin by $f\,\Delta p^2/p_H^2$.

**Answer (i).** $\mu(R) = 1 + f\big(\hat{B}^2(R)\big)\,(\Delta p^2/p_H^2)\,(p_H^2 R^2 - I^2) \;\geq 1$, with $\hat{B}^2(R) = (\Delta p^2/p_H^2)(p_H^2 R^2 - I^2 + R)$ and gross utility $W(R) = R + F(\hat{B}^2(R))(p_H^2 R^2 - I^2)$.

**Step 4 — (ii) Date-1 incentive constraint.** By (64) and Step 1, $W(0) = 0 + F(\hat{B}^2(0))\,NPV^2 = 0$, so the entrepreneur works at date 1 iff

$$\begin{aligned} p_H^1 W(R_b^1) + (1-p_H^1)W(0) &\geq p_L^1 W(R_b^1) + (1-p_L^1)W(0) + B^1 \\ (p_H^1 - p_L^1)W(R_b^1) - (p_H^1 - p_L^1)W(0) &\geq B^1 \\ (\Delta p^1)\,W(R_b^1) &\geq B^1 \qquad (W(0)=0) \\ (\Delta p^1)\big[R_b^1 + F(\hat{B}^2(R_b^1))NPV^2\big] &\geq B^1, \end{aligned} \tag{IC$^1$}$$

the last line substituting (64). Let $\tilde{R}_b^1$ be the smallest $R_b^1$ satisfying (IC$^1$). Whenever $F(\hat{B}^2(\tilde{R}_b^1)) > 0$,

$$\tilde{R}_b^1 = \frac{B^1}{\Delta p^1} - F(\hat{B}^2(\tilde{R}_b^1))NPV^2 < \frac{B^1}{\Delta p^1}. \tag{66}$$

**Step 5 — (ii) Date-1 financing.** The entrepreneur maximises $p_H^1 W(R_b^1) - A^1$ subject to (IC$^1$) and $p_H^1(R^1 - R_b^1) \geq I^1 - A^1$. Since $W$ is increasing (65), the breakeven constraint binds, and financing is possible iff

$$p_H^1(R^1 - \tilde{R}_b^1) \;\geq\; I^1 - A^1. \tag{67}$$

By (66) the left-hand side exceeds the stand-alone pledgeable income $p_H^1(R^1 - B^1/\Delta p^1)$. So for $A^1$ with $p_H^1(R^1 - B^1/\Delta p^1) < I^1 - A^1 \leq p_H^1(R^1 - \tilde{R}_b^1)$, project 1 is funded only because project 2 exists, although none of project 2's income is pledged at date 1.

**Step 6 — (ii) An example.** Project 2: $I^2 = 20$, $R^2 = 40$, $p_H^2 = 0.75$, $p_L^2 = 0.25$, $B^2 \sim U[7,9]$, so $NPV^2 = 10$ and $\Delta p^2 NPV^2/p_H^2 = 20/3 < 7$ (assumption holds; also $p_L^2 R^2 + B^2 < I^2$). By (63), $\hat{B}^2(R) = \frac{2}{3}(10 + R)$, which lies in $[7,9]$ for $R \in [0.5, 3.5]$; there $F = (\hat{B}^2 - 7)/2$ and (64)–(65) give

$$\begin{aligned} W(R) &= R + \frac{\frac{2}{3}(10+R) - 7}{2}\,(10) \\ &= R + \frac{(20 + 2R - 21)/3}{2}\,(10) = R + \frac{5(2R-1)}{3} \\ &= \frac{3R + 10R - 5}{3} = \frac{13R - 5}{3}, \\ \mu &= 1 + \Big(\frac{1}{2}\Big)\Big(\frac{0.5}{0.75}\Big)(10) = 1 + \frac{10}{3} = \frac{13}{3}. \end{aligned}$$

Project 1: $I^1 = 4.5$, $R^1 = 10$, $p_H^1 = 0.5$, $p_L^1 = 0.25$, $B^1 = 0.5$, so $B^1/\Delta p^1 = 2$. Alone, the pledgeable income is $(0.5)(10 - 2) = 4$ and $\bar{A}^1 = 4.5 - 4 = 0.5$. With project 2, (IC$^1$) at equality reads $W(\tilde{R}_b^1) = B^1/\Delta p^1$, i.e. $(13\tilde{R}_b^1 - 5)/3 = 2$, so $13\tilde{R}_b^1 = 6 + 5 = 11$, $\tilde{R}_b^1 = 11/13 \approx 0.846154$ (inside $[0.5, 3.5]$, where the formula for $W$ holds), and

$$p_H^1(R^1 - \tilde{R}_b^1) = (0.5)\Big(10 - \frac{11}{13}\Big) = \frac{119}{26} \approx 4.576923 > 4.5.$$

The threshold falls from 0.5 to $4.5 - 119/26 = -1/13 \approx -0.076923$: any $A^1 \in [0, 0.5)$ is now funded (at $A^1 = 0$, $R_b^1 = 10 - 4.5/0.5 = 1 \geq \tilde{R}_b^1$).

**Answer (ii).** Yes. Because project 2's rent is lost on failure, the date-1 reward needed for effort falls to $\tilde{R}_b^1 < B^1/\Delta p^1$ and date-1 pledgeable income rises to $p_H^1(R^1 - \tilde{R}_b^1) > p_H^1(R^1 - B^1/\Delta p^1)$. Any $A^1$ with $p_H^1(R^1 - B^1/\Delta p^1) < I^1 - A^1 \leq p_H^1(R^1 - \tilde{R}_b^1)$ is financed only thanks to project 2 (in the example, $A^1 \in [0, 0.5)$).

**Intuition.** Retained earnings are worth more than their face value because they are the net worth that gets the next project financed. That future rent is lost if project 1 fails, so it acts as a reward for date-1 effort: part of the incentive is paid with continuation value instead of cash, and the cash saved becomes pledgeable to date-1 lenders.

**Common mistakes.**

- Valuing retained earnings at face ($\mu = 1$). Each unit also raises the chance that project 2 is financed, worth $f\,(\Delta p^2/p_H^2)\,NPV^2$ more.
- Writing the date-1 incentive constraint with $R_b^1$ alone, $(\Delta p^1)R_b^1 \geq B^1$: the reward for success is $W(R_b^1)$, which includes the continuation rent.
- Treating project 2's income as pledgeable at date 1. The effect works only through the entrepreneur's incentives; date-1 lenders are repaid out of $R^1$.

**Book answer.** Matches the book's answer (p. 575): $\mu = 1 + f(\hat{B}^2)(d\hat{B}^2/dR_b^1)NPV^2$ and the date-1 constraint $(\Delta p^1)[R_b^1 + F(\hat{B}^2(R_b^1))NPV^2] \geq B^1$. The book writes the constraint without $W(0)$; this is right because the assumption on $\underline{B}^2$ makes $W(0) = 0$ (Step 1). The numerical example is this solution's.

**Ref:** Tirole §3.2.2, pp. 116–117; §3.4.2, p. 128 (shadow value of equity); answer p. 575.

## 12. Renegotiation and debt forgiveness — Tirole Exercise 3.6, p. 146

**What is asked.** In the variable-investment model, suppose the initial loan leaves the borrower too small a stake to work, $R_b < BI/\Delta p$, but can be renegotiated after the investment and before effort. (i) Show that a mutually beneficial renegotiation exists exactly when the stated inequality holds. (ii) Explain that inequality, and derive it as the financing condition of a fictitious fixed-investment project. (iii) If the borrower makes a take-it-or-leave-it renegotiation offer, find how much she can invest (her borrowing capacity) with such an initial contract. (iv) Argue directly that writing $R_b \geq BI/\Delta p$ from the start loses nothing.

**Approach.** Lecture Topic 4B, renegotiation. Without renegotiation the borrower shirks; renegotiation can only help if it restores ($\text{IC}_b$), and it happens iff both sides gain relative to the shirking status quo. The lender gives up part of his claim (*debt forgiveness*) and is paid back out of the extra *pledgeable income* that effort creates, so the condition has the form "pledgeable income $\geq$ investment $-$ cash" of the fixed-investment model. For (iii) the lender's ex ante breakeven uses his renegotiated payoff; for (iv) rational expectations make the renegotiated contract one that could have been signed directly.

**Notation.**

- $I$: (sunk) investment; income $RI$ in success, 0 in failure; $A$: borrower's cash.
- $p_H$, $p_L = p_H - \Delta p$; private benefit $BI$ from shirking; (3.7) $p_H R > 1$, (3.8) $1 > p_L R + B$, (3.9) $p_H R < 1 + p_H B/\Delta p$; these imply $(\Delta p)R > B$.
- $R_b < BI/\Delta p$: borrower's stake in the initial contract; $\hat{R}_b$: renegotiated stake.
- $k = 1/[1 - p_H(R - B/\Delta p)]$: equity multiplier (3.12).

**Step 1 — (i) Status quo.** We compute each party's payoff if nothing is renegotiated; these are the outside options in the renegotiation. With $R_b < BI/\Delta p$, $(\Delta p)R_b < BI$: ($\text{IC}_b$) fails and the borrower shirks (success probability $p_L$, private benefit $BI$). Payoffs if the contract is not renegotiated:

$$U_b^0 = p_L R_b + BI, \tag{68}$$

$$U_l^0 = p_L(RI - R_b). \tag{69}$$

**Step 2 — (i) Renegotiation must restore incentives.** A new stake $\hat{R}_b < BI/\Delta p$ still induces shirking, so total surplus $p_L RI + BI$ is unchanged and the change is a pure transfer: one side loses.

A mutually advantageous renegotiation therefore needs $\hat{R}_b \geq BI/\Delta p$, and both parties must gain relative to (68)–(69):

$$p_H \hat{R}_b \geq p_L R_b + BI, \tag{70}$$

$$p_H(RI - \hat{R}_b) \geq p_L(RI - R_b). \tag{71}$$

**Step 3 — (i) The condition.** For any $\hat{R}_b \geq BI/\Delta p$, (70) holds automatically, since $R_b < BI/\Delta p$:

$$p_H \hat{R}_b \geq p_H \frac{BI}{\Delta p} = BI + p_L \frac{BI}{\Delta p} > BI + p_L R_b.$$

The lender prefers the smallest admissible stake, $\hat{R}_b = BI/\Delta p$. So a mutually advantageous $\hat{R}_b$ exists iff (71) holds at $\hat{R}_b = BI/\Delta p$:

$$\begin{aligned} p_H\Big(RI - \frac{BI}{\Delta p}\Big) &\geq p_L(RI - R_b) \\ p_H RI - \frac{p_H BI}{\Delta p} &\geq p_L RI - p_L R_b \\ (p_H - p_L)RI - \frac{p_H BI}{\Delta p} + p_L R_b &\geq 0 \\ (\Delta p)RI - \frac{p_H BI}{\Delta p} + p_L R_b &\geq 0. \end{aligned} \tag{72}$$

Necessity: any renegotiated contract has $\hat{R}_b \geq BI/\Delta p$, and (71) is hardest to meet at larger $\hat{R}_b$. Sufficiency: if (72) holds, $\hat{R}_b = BI/\Delta p$ makes the lender weakly and the borrower strictly better off. The renegotiation is *debt forgiveness*: the lender's claim in success falls from $RI - R_b$ to $RI - \hat{R}_b$.

**Answer (i).** Renegotiation (to $\hat{R}_b \geq BI/\Delta p$) takes place iff $(\Delta p)RI - p_H BI/\Delta p + p_L R_b \geq 0$.

**Step 4 — (ii) Interpretation via the general theory.** Treat "inducing effort" as a fictitious fixed-investment project. Its income is the increase in expected output, $(\Delta p)RI$; it needs no new investment ($I_f = 0$); the borrower's minimum stake in it is $p_H\, BI/\Delta p$ (her expected rent under ($\text{IC}_b$)), and she brings as "cash" the expected income she gives up, $A_f \equiv p_L R_b$. Its pledgeable income is $(\Delta p)RI - p_H BI/\Delta p$, and the financing condition ($\text{IR}_l$), "pledgeable income $\geq$ investment $-$ cash", reads

$$(\Delta p)RI - \frac{p_H BI}{\Delta p} \;\geq\; 0 - p_L R_b, \tag{73}$$

which is (72). The total gain from restoring effort, $(\Delta p)RI - BI$, is always positive, yet renegotiation may fail: the lender can only be paid out of pledgeable income, and the borrower's initial stake $p_L R_b$ plays the role of net worth. A larger $R_b$ makes forgiveness easier.

**Answer (ii).** The condition says the pledgeable part of the effort gain, $(\Delta p)RI - p_H BI/\Delta p$, must cover the lender's "outlay" $0 - p_L R_b$: it is ($\text{IR}_l$) for a fictitious project with income $(\Delta p)RI$, investment 0 and cash $p_L R_b$.

**Step 5 — (iii) Borrowing capacity under renegotiation, borrower has all bargaining power.** When (72) holds, the borrower offers the $\hat{R}_b$ that leaves the lender indifferent:

$$p_H(RI - \hat{R}_b) = p_L(RI - R_b), \quad \text{i.e.} \quad \hat{R}_b = RI - \frac{p_L}{p_H}(RI - R_b). \tag{74}$$

The lender therefore anticipates exactly (69), and his ex ante breakeven constraint is

$$p_L(RI - R_b) \;\geq\; I - A. \tag{75}$$

The borrower's utility is total surplus with effort minus the lender's payoff: $p_H RI - p_L(RI - R_b) - A$, which by the binding (75) equals $p_H RI - (I - A) - A = (p_H R - 1)I$, the NPV with effort. Since

$p_H R > 1$ (3.7), she maximises $I$ subject to (72), (75) and $0 \le R_b < BI/\Delta p$. (72) bounds $p_L R_b$ from below and (75) (expanded: $p_L RI - p_L R_b \ge I - A$) bounds it from above:

$$\frac{p_H BI}{\Delta p} - (\Delta p)RI \;\le\; p_L R_b \;\le\; p_L RI - (I - A). \tag{76}$$

An $R_b$ exists iff the lower bound is below the upper bound:

$$\begin{aligned}
\frac{p_H BI}{\Delta p} - (\Delta p)RI &\le p_L RI - I + A \\
I - (\Delta p)RI - p_L RI + \frac{p_H BI}{\Delta p} &\le A \\
I - p_H RI + \frac{p_H BI}{\Delta p} &\le A \qquad (\Delta p + p_L = p_H) \\
I\Big[1 - p_H\Big(R - \frac{B}{\Delta p}\Big)\Big] &\le A \\
I &\le kA.
\end{aligned} \tag{77}$$

If $p_H B/\Delta p \ge (\Delta p)R$, the lower bound in (76) is nonnegative and the capacity is exactly $kA$: at $I = kA$ both (72) and (75) bind, $R_b < BI/\Delta p$ (by $(\Delta p)R > B$) and (74) gives $\hat{R}_b = BI/\Delta p$. If $p_H B/\Delta p < (\Delta p)R$, the constraint $R_b \ge 0$ binds first, and (75) with $R_b = 0$ gives $I \le A/(1 - p_L R) < kA$. Either way the capacity is at most $kA$, the capacity without renegotiation.

**Answer (iii).** The borrowing capacity is $I = kA$, $k = 1/[1 - p_H(R - B/\Delta p)]$, when $p_H B/\Delta p \ge (\Delta p)R$ (then $p_L R_b = kA[p_H B/\Delta p - (\Delta p)R]$ and $\hat{R}_b = BI/\Delta p$), and $I = A/(1 - p_L R) < kA$ otherwise; it never exceeds the capacity $kA$ of an incentive-compatible initial contract.

**Step 6 — (iv) Rational expectations.** Take any initial contract $(I, R_b)$ with $R_b < BI/\Delta p$. If it is not renegotiated, the borrower shirks; the NPV is $(p_L R + B - 1)I < 0$ by (3.8), and since the lender breaks even the borrower's utility is that negative NPV: worse than not investing. If it is renegotiated to $\hat{R}_b \ge BI/\Delta p$, the final contract $(I, \hat{R}_b)$ satisfies ($\text{IC}_b$), and by rational expectations the lender's ex ante breakeven uses his final payoff, $p_H(RI - \hat{R}_b) \ge I - A$, which is ($\text{IR}_l$). So $(I, \hat{R}_b)$ could have been signed directly, with the same effort and the same payoffs. Nothing is lost by restricting attention to $R_b \ge BI/\Delta p$, which is renegotiation-proof.

**Answer (iv).** Any initial contract with $R_b < BI/\Delta p$ either is not renegotiated (negative NPV, worse than not investing) or ends as an incentive-compatible contract that satisfies ($\text{IR}_l$) and could have been signed at the outset; so restricting to $R_b \ge BI/\Delta p$ loses nothing.

**Intuition.** A lender facing a borrower who is about to shirk is willing to forgive debt, because the higher stake raises the success probability and the lender keeps part of that gain. He forgives only if his remaining claim after forgiveness, the pledgeable part of the effort gain, beats what he gets from a shirking borrower. Anticipating this, an initial contract that relies on forgiveness can at best reproduce the incentive-compatible contract: the multiplier $k$ is the right one.

**Common mistakes.**

- Arguing that renegotiation always happens because restoring effort raises total surplus ($(\Delta p)RI > BI$). The lender gains only the *pledgeable* part of that surplus.
- Writing the lender's ex ante breakeven in (iii) as $p_H(RI - R_b) \ge I - A$, i.e. his initial claim with effort. He anticipates the renegotiation and values his final payoff, which with a take-it-or-leave-it offer by the borrower is $p_L(RI - R_b)$.
- Mixing the fixed- and variable-investment notation: here the private benefit is $BI$ and the minimum stake is $BI/\Delta p$, not $B/\Delta p$.

**Book answer.** Matches the book's answer to (i)–(ii) (pp. 573–574), including the fictitious-project reading with cash $\hat{A} = p_L R_b$. The book gives no answer to (iii)–(iv); Steps 5–6 are this solution's.

**Ref:** Tirole §3.4.1–3.4.2, pp. 127–128; §3.2.2, pp. 116–117; answer pp. 573–574.

## 13. Loan size and collateral requirements — Tirole Exercise 4.17, p. 195

**What is asked.** The project has decreasing returns, $R(I)$ concave, and the entrepreneur can pledge costly collateral $C$ that investors seize on failure. Assuming the first best cannot be financed, show that a poorer entrepreneur (lower $A$) or a worse agency problem (higher $B$, or higher $p_H/\Delta p$ with $p_H$ fixed) leads to a smaller optimal investment and to more collateral.

**Approach.** This combines the variable-investment model (§3.4) with collateral pledging (§4.3.5); lecture Topic 4A, collateral reading. Collateral raises *pledgeable income* twice: investors get $\phi(C)$ on failure, and the borrower's stake grows, so her cash reward can shrink by $C$ ($p_H C$). It costs a deadweight loss $C-\phi(C)$ on failure. We eliminate $R_b$ with the binding ($\text{IC}_b$), write the Lagrangian with multiplier $\mu$ on ($\text{IR}_l$), derive the first-order conditions, and do comparative statics on the resulting system with the implicit-function theorem.

**Notation.**

- $I \geq 0$: investment; $R(I)$: income in success ($R(0)=0$, $R'>0$, $R''<0$, Inada).
- $p_H$, $p_L = p_H - \Delta p$; private benefit $BI$; $b \equiv p_H B/\Delta p$: per-unit agency cost (rises with $B$ and, with $p_H$ fixed, with $p_H/\Delta p$).
- $C \geq 0$: collateral pledged in failure, cost $C$ to the entrepreneur, value $\phi(C)$ to investors ($\phi(0)=0$, $\phi'>0$, $\phi''<0$, $\phi'(0)=1$, $\phi'(\infty)=0$).
- $R_b \geq 0$: reward in success; $A$: cash; $I^*$: first-best size, $p_H R'(I^*)=1$.
- $\mu \geq 0$: multiplier of the breakeven constraint; $\theta \equiv \mu/(1+\mu) \in [0,1)$; $\varepsilon(I) \equiv -IR''(I)/R'(I) > 0$: curvature of $R$.

**Assumption.** The limited-liability constraint $R_b \geq 0$, i.e. $C \leq BI/\Delta p$, does not bind (as in footnote 24 of §4.3.5).

**Step 1 — Programme.** We set up the problem and eliminate $R_b$. The entrepreneur gets $R_b$ on success and $-C$ on failure, so ($\text{IC}_b$) is $p_H R_b - (1-p_H)C \geq p_L R_b - (1-p_L)C + BI$, i.e. $(\Delta p)R_b + (\Delta p)C \geq BI$, i.e. $(\Delta p)(R_b + C) \geq BI$. Investors get $R(I) - R_b$ in success and $\phi(C)$ in failure. Collateral destroys $C - \phi(C) \geq 0$ in failure, so with binding breakeven her utility is the NPV:

$$\max_{I,C,R_b} U_b = p_H R(I) - I - (1-p_H)\big[C - \phi(C)\big] \tag{78}$$

$$\text{s.t.} \quad (\Delta p)(R_b + C) \;\geq\; BI, \tag{$\text{IC}_b$}$$

$$p_H\big[R(I) - R_b\big] + (1-p_H)\phi(C) \;\geq\; I - A. \tag{$\text{IR}_l$}$$

$R_b$ does not enter (78) and lowers ($\text{IR}_l$), so ($\text{IC}_b$) binds: $R_b = BI/\Delta p - C$. Substituting into ($\text{IR}_l$),

$$p_H\Big[R(I) - \frac{BI}{\Delta p} + C\Big] + (1-p_H)\phi(C) \geq I - A$$

$$p_H R(I) - \frac{p_H B}{\Delta p} I + p_H C + (1-p_H)\phi(C) - I + A \geq 0$$

$$G(I,C) \equiv p_H R(I) - bI + p_H C + (1-p_H)\phi(C) - I + A \;\geq\; 0. \tag{79}$$

Collateral raises pledgeable income directly ($\phi$) and by replacing reward ($p_H C$). The objective is concave and $G$ is concave in $(I,C)$, so the Kuhn–Tucker conditions are sufficient.

**Step 2 — First-order conditions.** We derive the conditions that characterise the optimum. With multiplier $\mu \geq 0$ on (79), the Lagrangian is

$$\begin{aligned}\mathcal{L}(I,C,\mu) = {} & p_H R(I) - I - (1-p_H)\big[C - \phi(C)\big] \\ & + \mu\big[p_H R(I) - bI + p_H C + (1-p_H)\phi(C) - I + A\big].\end{aligned}$$

Differentiating with respect to $I$ and $C$ and setting the derivatives to zero,

$$\frac{\partial \mathcal{L}}{\partial I} = p_H R'(I) - 1 + \mu\big[p_H R'(I) - b - 1\big] = 0,$$
$$\frac{\partial \mathcal{L}}{\partial C} = -(1-p_H)\big[1-\phi'(C)\big] + \mu\big[p_H + (1-p_H)\phi'(C)\big] = 0.$$

*The $I$-condition.* Collect the terms in $p_H R'(I) - 1$:

$$(1+\mu)\big[p_H R'(I) - 1\big] - \mu b = 0$$
$$p_H R'(I) - 1 = \frac{\mu}{1+\mu}\, b.$$

*The $C$-condition.* Expand and collect the terms in $\phi'(C)$:

$$-(1-p_H) + (1-p_H)\phi'(C) + \mu p_H + \mu(1-p_H)\phi'(C) = 0$$
$$(1+\mu)(1-p_H)\phi'(C) = (1-p_H) - \mu p_H$$
$$\phi'(C) = \frac{1}{1+\mu} - \frac{\mu}{1+\mu}\,\frac{p_H}{1-p_H}.$$

Writing $\theta = \mu/(1+\mu)$, so that $1/(1+\mu) = 1-\theta$,

$$p_H R'(I) - 1 = \theta\, b, \tag{80}$$
$$\phi'(C) = 1 - \theta - \theta\frac{p_H}{1-p_H} = 1 - \theta\,\frac{(1-p_H)+p_H}{1-p_H} = 1 - \frac{\theta}{1-p_H}. \tag{81}$$

If $\mu = 0$, (80)–(81) give $I = I^*$, $C = 0$, which violates (79) by the assumption $p_H[R(I^*) - BI^*/\Delta p] < I^* - A$. So $\theta > 0$: (80) gives $I < I^*$ and (81) gives $\phi'(C) < 1 = \phi'(0)$, i.e. $C > 0$; finiteness of $C$ needs $\theta < 1 - p_H$. These are the book's conditions, since $\theta = \mu/(1+\mu)$.

**Step 3 — Monotonicity in $\theta$.** The FOCs define $I$ and $C$ as functions of $\theta$ (and $b$); we find how they move and then pin $\theta$ down with the binding breakeven. Totally differentiating (80), $p_H R''(I)\,dI = b\,d\theta + \theta\,db$; and (81), $\phi''(C)\,dC = -d\theta/(1-p_H)$. With $R'' < 0$, $\phi'' < 0$ and $Q \equiv 1/(p_H|R''(I)|) > 0$, dividing by $p_H R'' = -1/Q$ and by $\phi''$:

$$\frac{\partial I}{\partial \theta} = -bQ < 0, \qquad \frac{\partial I}{\partial b} = -\theta Q < 0, \tag{82}$$
$$C'(\theta) = -\frac{1}{(1-p_H)\phi''(C)} > 0. \tag{83}$$

Let $g(\theta; A, b) \equiv G(I(\theta, b), C(\theta))$, the slack of (79) along the FOCs; the optimum is the root $g = 0$. Using (80), $p_H R' = 1 + \theta b$, so $G_I = p_H R' - b - 1 = (1+\theta b) - b - 1 = -(1-\theta)b < 0$; also $G_C = p_H + (1-p_H)\phi' > 0$. Then, by the chain rule and (82)–(83),

$$\begin{aligned} g_\theta &= G_I \frac{\partial I}{\partial \theta} + G_C C'(\theta) \\ &= \big(-(1-\theta)b\big)\big(-bQ\big) + G_C C'(\theta) \\ &= (1-\theta)b^2 Q + G_C C'(\theta) \;> 0. \end{aligned} \tag{84}$$

$g$ is $< 0$ at $\theta = 0$ (first best not pledgeable) and $\to +\infty$ as $\theta \to 1 - p_H$ ($C \to \infty$), so the root $\theta^* \in (0, 1-p_H)$ is unique.

**Step 4 — Comparative statics in $A$.** $g_A = 1$, so by the implicit-function theorem and (84)

$$\frac{d\theta^*}{dA} = -\frac{g_A}{g_\theta} = -\frac{1}{g_\theta} < 0. \tag{85}$$

Combining with (82)–(83) by the chain rule:

$$\frac{dI}{dA} = \frac{\partial I}{\partial \theta}\,\frac{d\theta^*}{dA} = (-bQ)\Big(-\frac{1}{g_\theta}\Big) = \frac{bQ}{g_\theta} > 0,$$
$$\frac{dC}{dA} = C'(\theta)\,\frac{d\theta^*}{dA} = -\frac{C'(\theta)}{g_\theta} < 0.$$

As $A$ decreases, $\theta^*$ (hence $\mu = \theta/(1-\theta)$) rises, the investment size falls and the collateral rises, as claimed.

**Answer (balance sheet).** As $A$ decreases, the shadow cost $\mu$ rises, investment $I$ falls and collateral $C$ rises: $dI/dA > 0$, $dC/dA < 0$.

**Step 5 — Comparative statics in the agency cost $b$.** Both $B$ and $p_H/\Delta p$ (with $p_H$ fixed) enter only through $b = p_H B/\Delta p$, so it suffices to vary $b$. Now $b$ also enters $I(\theta, b)$; by (79), $G_b = -I$, so

$$\begin{aligned} g_b &= G_b + G_I\frac{\partial I}{\partial b} \\ &= -I + \big(-(1-\theta)b\big)\big(-\theta Q\big) = -I + \theta(1-\theta)bQ, \end{aligned} \tag{86}$$

and by the implicit-function theorem $d\theta^*/db = -g_b/g_\theta$. *Investment.* Using (82), (84) and (86) (with $D \equiv G_C C' > 0$),

$$\begin{aligned} \frac{dI}{db} &= \frac{\partial I}{\partial \theta}\frac{d\theta^*}{db} + \frac{\partial I}{\partial b} = (-bQ)\Big(-\frac{g_b}{g_\theta}\Big) + (-\theta Q) = -Q\Big(\theta - b\,\frac{g_b}{g_\theta}\Big) \\ &= -Q\,\frac{\theta g_\theta - b g_b}{g_\theta} = -Q\,\frac{\theta D + bI}{g_\theta} \;< 0, \end{aligned} \tag{87}$$

where the numerator is $\theta g_\theta - b g_b = \theta(1-\theta)b^2Q + \theta D + bI - \theta(1-\theta)b^2Q = \theta D + bI > 0$. So $I$ falls with $B$ and with $p_H/\Delta p$, as claimed. *Collateral.* $dC/db = C'(\theta)\,d\theta^*/db = -C'(\theta)g_b/g_\theta$ has the sign of $-g_b$, and $-g_b > 0$ iff $I > \theta(1-\theta)bQ$, i.e. (multiplying by $1/Q = p_H|R''|$) iff $p_H|R''|I > \theta(1-\theta)b$. By the definition of $\varepsilon$ and (80), $p_H|R''|I = \varepsilon(I)\,p_H R'(I) = \varepsilon(I)(1+\theta b)$, so

$$\frac{dC}{db} > 0 \iff \varepsilon(I)\,(1+\theta b) \;>\; \theta(1-\theta)\,b. \tag{88}$$

This holds when $R$ is concave enough (for instance $\varepsilon \geq 1$, since $\theta(1-\theta)b < 1 + \theta b$), but it can fail when $R$ is close to linear. Counterexample: $R(I) = 3I^{0.95}$ ($\varepsilon = 0.05$), $\phi(C) = \ln(1+C)$, $p_H = 0.5$, $\Delta p = 0.25$, $A = 1$. For $B = 0.5$ ($b = 1$) the solution is $\theta^* = 0.3276$, $I = 4.1203$, $C = 1.9003$; for $B = 2$ ($b = 4$) it is $\theta^* = 0.1238$, $I = 0.3815$, $C = 0.3292$. Collateral *falls* as the agency cost rises; at $b = 1$, (88) reads $0.05(1 + 0.3276) = 0.0664 > 0.3276(1 - 0.3276)(1) = 0.2203$, false. (The first best is not pledgeable in both cases and $C \leq BI/\Delta p$.)

**Answer (agency cost).** Investment always falls with the agency cost, $dI/db < 0$. Collateral rises with it iff $\varepsilon(I)(1+\theta b) > \theta(1-\theta)b$ (e.g. whenever $\varepsilon \geq 1$); the statement's claim for $C$ is not general, as the counterexample shows.

**Intuition.** A weaker balance sheet tightens the financing constraint; the entrepreneur pays for funding on both margins she controls, investing less (which saves outside funds) and pledging more collateral (which adds pledgeable income at a deadweight cost). A higher agency cost also lowers investment, but it has a second effect: each unit of investment now absorbs more pledgeable income, so the downsizing itself releases funds. When output is nearly linear in $I$, downsizing is cheap and strong enough that less collateral is needed. Only the balance-sheet comparative static is unambiguous.

**Common mistakes.**

- Dropping the indirect effect of collateral, the term $p_H C$: collateral also lets the cash reward $R_b$ fall by $C$ without breaking (IC$_b$).

- Treating $b$ as a pure shift of the constraint. It also enters the $I$-condition (80), which is why the sign of $dC/db$ is ambiguous.
- Ignoring limited liability, $R_b = BI/\Delta p - C \geq 0$: the substitution of collateral for reward is valid only while it holds.

**Book answer.** Agrees with the book's answer (p. 591): the FOCs (80)–(81) and "as the balance sheet deteriorates, $\mu$ increases, $I$ decreases, and $C$ increases". The book's answer does not claim that collateral rises with the agency cost (it says only that borrowing rises when the agency cost falls). The statement's claim for $C$ holds only under (88); the claim for $I$ holds in general (87).

**Ref:** Tirole §4.3.5, p. 170; §3.4.2, pp. 127–128; answer p. 591.

## 14. Dealing with asset substitution — Tirole Exercise 7.3, pp. 323–324

**What is asked.** (i) Without asset substitution: explain the threshold equation (1), give the entrepreneur's expected utility, and describe the optimal contracts (at least at $A = \bar{A}$). (ii) Now she can secretly run down maintenance (lower liquidation value $L$) to raise the success probability. Explain condition (2), find the smallest $A$ for which this threat does no harm, and describe the contract that achieves it.

**Approach.** The fixed-investment model with a verifiable liquidation value (lecture Topic 4B, asset substitution and remedies). Liquidation value is not subject to moral hazard, so all of it is pledgeable; continuation income is pledgeable only up to $R - B/\Delta p$. A debt-like claim pays the entrepreneur only in continuation, so she prefers to shift value from liquidation to continuation: *asset substitution*. To prevent it she must be paid something in liquidation, conditional on good maintenance; that payment is a rent, which lowers *pledgeable income* and raises the minimum equity. We write ($\text{IR}_l$), ($\text{IC}_b$) and a no-substitution constraint, and minimise her rent.

**Notation.**

- $I > A$: fixed investment; $A$: cash; $x \in (0,1)$: probability of demand; with probability $1-x$ there is no demand and the assets are liquidated for $L$.
- $R$: income in success; $p_H$, $p_L = p_H - \Delta p$, $B$: as in Section 3.2.
- $L_0$: liquidation value without asset substitution; $\tau(L)$: change in success probability when maintenance yields liquidation value $L$ ($\tau' < 0$, $\tau'' < 0$, $\tau(L_0) = 0$).
- $r_b$: entrepreneur's payment on liquidation; $R_b$: payment on continuation and success.
- $U_b^*$: entrepreneur's utility; $\bar{A}$: threshold (1); $A^*$: threshold with asset substitution.

**Assumption.** The statement's "(hence, with probability $x$)" for liquidation is read as probability $1-x$, the probability of no demand. $L \geq 0$ is chosen from $[0, \infty)$.

**Step 1 — (i) Interpreting (1).** The liquidation value $L_0$ is verifiable and not subject to moral hazard, so all of it can go to investors. On continuation only $R - R_b$ with $R_b \geq B/\Delta p$ can. The left-hand side of (1) is the maximal expected pledgeable income,

$$\mathcal{P}_0 = (1-x)L_0 + xp_H\Big(R - \frac{B}{\Delta p}\Big), \tag{89}$$

and (1) sets it equal to the investors' outlay at the threshold: financing is feasible iff $A \geq \bar{A} = I - \mathcal{P}_0$. The side condition $L_0 \leq p_H(R - B/\Delta p)$ says that, when there is demand, investors also prefer continuation (their maximal continuation claim beats $L_0$): there is no conflict over the continuation decision.

**Answer (i), interpret (1).** The left side of (1) is the maximal expected pledgeable income (all of $L_0$ on liquidation, $R - B/\Delta p$ on success after continuation); (1) defines $\bar{A}$ as the cash at which it just covers the investors' outlay $I - A$.

**Step 2 — (i) Utility and optimal contracts.** Competitive investors break even, so the entrepreneur receives the expected value created minus the investment, net of her own cash: $U_b^* = [(1-x)L_0 + xp_H R] - (I - A) - A$, i.e.

$$U_b^* = (1-x)L_0 + xp_H R - I. \tag{90}$$

A contract $(r_b, R_b)$, liquidating iff there is no demand, is optimal iff

$$(1-x)(L_0 - r_b) + xp_H(R - R_b) = I - A, \tag{91}$$

$$R_b \geq \frac{B}{\Delta p}, \quad r_b \geq 0. \tag{92}$$

For $A > \bar{A}$ there is a continuum (the slack $A - \bar{A}$ can be paid as $r_b$ or as extra $R_b$). At $A = \bar{A}$, subtracting (1), $I - \bar{A} = (1-x)L_0 + xp_H(R - B/\Delta p)$, from (91),

$$-(1-x)\, r_b - xp_H\Big(R_b - \frac{B}{\Delta p}\Big) = 0,$$

and both terms are $\leq 0$ by (92), so each is zero: $r_b = 0$ and $R_b = B/\Delta p$. This is a debt contract with face value $D = R - B/\Delta p \geq L_0$ (since $L_0 \leq p_H(R - B/\Delta p) \leq D$). On liquidation the firm defaults and investors take $L_0$; on success they are repaid $D$.

**Answer (i), utility and contracts.** $U_b^* = (1-x)L_0 + xp_H R - I$. Optimal contracts: any $(r_b, R_b)$ with $r_b \geq 0$, $R_b \geq B/\Delta p$ and binding (91); at $A = \bar{A}$ the unique one is debt with face $D = R - B/\Delta p$ ($r_b = 0$, $R_b = B/\Delta p$).

**Step 3 — (ii) Interpreting (2).** With maintenance level $L$ (and effort), the NPV is

$$NPV(L) = (1-x)L + x\big[p_H + \tau(L)\big]R - I. \tag{93}$$

Differentiating, $NPV'(L) = (1-x) + x\tau'(L)R$ and $NPV''(L) = x\tau''(L)R < 0$, so it is strictly concave, and its first-order condition is $(1-x) + x\tau'(L)R = 0$, i.e. (dividing by $x$) $\tau'(L)R = -(1-x)/x$, which (2) says holds at $L_0$. So $L_0$ is the unique efficient maintenance level: one more unit of liquidation value, worth $1 - x$ in expectation, costs exactly $x|\tau'|R$ of expected continuation income. $\tau(L_0) = 0$ normalises the success probabilities so that $NPV(L_0) = U_b^*$.

**Answer (ii), interpret (2).** (2) says $L_0$ is the efficient maintenance level: it maximises the NPV, and at $L_0$ the success probabilities are those of part (i).

**Step 4 — (ii) The entrepreneur's maintenance choice.** Under the step contract ($r_b(L) = r_b$ if $L \geq L_0$, 0 otherwise) and anticipating effort at stage 4 (her (IC$_b$), $(\Delta p)R_b \geq B$, does not depend on $L$), she chooses $L$ to maximise

$$U(L) = (1-x)\, r_b(L) + x\big[p_H + \tau(L)\big]R_b. \tag{94}$$

For $L < L_0$ she gets 0 on liquidation and $\tau$ is decreasing, so the best deviation is $L = 0$; any $L > L_0$ lowers $\tau$ for no gain. So $L_0$ is chosen iff $U(L_0) \geq U(0)$:

$$\begin{aligned}(1-x)r_b + x\big[p_H + \tau(L_0)\big]R_b &\geq (1-x)(0) + x\big[p_H + \tau(0)\big]R_b \\ (1-x)r_b + xp_H R_b &\geq xp_H R_b + x\tau(0)R_b \qquad (\tau(L_0) = 0) \\ (1-x)\, r_b &\geq x\,\tau(0)\, R_b. \end{aligned} \tag{NS}$$

A debt contract ($r_b = 0$) violates (NS): the entrepreneur would run the assets down to gamble on continuation.

**Step 5 — (ii) Minimum $A$.** Financing at the efficient $L_0$ requires the investors' breakeven (91) with (IC$_b$) and (NS). The entrepreneur's expected rent $(1-x)r_b + xp_H R_b$ is minimised by (NS) at equality (any larger $r_b$ is pure rent); substituting $(1-x)r_b = x\tau(0)R_b$, the rent is

$x\tau(0)R_b + xp_H R_b = x\big[p_H + \tau(0)\big]R_b$, increasing in $R_b$, so $R_b = B/\Delta p$, the lowest value ($\text{IC}_b$) allows. Then

$$r_b = \frac{x\,\tau(0)}{1-x}\,\frac{B}{\Delta p}. \tag{95}$$

Substituting $R_b = B/\Delta p$ and (95) into (91) at equality,

$$\begin{aligned} I - A^* &= (1-x)(L_0 - r_b) + xp_H\Big(R - \frac{B}{\Delta p}\Big) \\ A^* &= \Big[I - (1-x)L_0 - xp_H\Big(R - \frac{B}{\Delta p}\Big)\Big] + (1-x)\,r_b \\ &= \bar{A} + (1-x)\,r_b = \bar{A} + x\,\tau(0)\,\frac{B}{\Delta p}, \end{aligned} \tag{96}$$

using (1) for the bracket and (95) for $(1-x)r_b$. For $A \geq A^*$, asset substitution is innocuous: the firm is financed, maintains $L_0$ and the entrepreneur gets $U_b^*$. For $A \in [\bar{A}, A^*)$ the efficient contract is not fundable. Illustration: $x = 0.6$, $p_H = 0.8$, $p_L = 0.4$, $B = 1$, $R = 10$, $L_0 = 3$, $I = 5.5$, $\tau(L) = (9 - L^2)/90$ (which satisfies (2): $\tau'(3)R = -(6/90)(10) = -2/3 = -(0.4/0.6)$): $U_b^* = 0.4(3)+0.6(0.8)(10)-5.5 = 1.2+4.8-5.5 = 0.5$, $B/\Delta p = 2.5$, $\bar{A} = 5.5-1.2-0.6(0.8)(10-2.5) = 5.5-1.2-3.6 = 0.7$, $\tau(0) = 9/90 = 0.1$, $r_b = (0.6)(0.1)(2.5)/(0.4) = 0.375$, $A^* = 0.7 + (0.4)(0.375) = 0.85$.

**Step 6 — (ii) The optimal contract.** It is debt plus a reward for good maintenance: the entrepreneur keeps $B/\Delta p$ on success and receives $r_b$ on liquidation, but only if the assets are in good condition (a severance payment conditional on $L \geq L_0$). Giving her a stake in the liquidation value counters her bias toward the continuation state, where her debt-like claim pays.

**Answer (ii), minimum $A$ and contract.** $A^* = \bar{A} + x\,\tau(0)\,B/\Delta p$. The contract is debt ($R_b = B/\Delta p$ on success) plus a payment $r_b = x\tau(0)B/[(1-x)\Delta p]$ on liquidation, made only if the assets were maintained at $L_0$.

**Intuition.** A claim that pays only in success makes the entrepreneur care only about continuation profit, so she cuts maintenance to raise the success probability: risk shifting at the expense of investors, who own the liquidation value. To stop it, she must share in the liquidation value. That share is an extra rent, so it lowers pledgeable income and raises the equity needed, by the expected gain from the most aggressive substitution, $x\,\tau(0)B/\Delta p$.

**Common mistakes.**

- Reading "liquidation … (hence, with probability $x$)" literally; liquidation happens when there is no demand, with probability $1-x$ (Figure 7.16).
- Keeping the debt contract of part (i) in part (ii). With $r_b = 0$ the no-substitution constraint fails and she runs the assets down to $L = 0$.
- Paying $r_b$ on liquidation unconditionally. A payment that does not depend on $L$ leaves her maintenance incentive unchanged; it must be conditional on $L \geq L_0$.

**Book answer.** Matches the book's answer (p. 602): the same interpretation of (1), $U_b^*$, the debt contract at $A = \bar{A}$, the step-function contract with $R_b = B/\Delta p$, the no-substitution inequality $(1-x)r_b + xp_H B/\Delta p \geq x[p_H + \tau(0)]B/\Delta p$ and the threshold $I - A^* = (1-x)(L_0 - r_b) + xp_H(R - B/\Delta p)$. The statement's "with probability $x$" for liquidation looks like a typo for $1-x$.

**Ref:** Tirole §3.2.2, pp. 116–117; §7.2.3 "Effort and Risk Taking", p. 314; answer p. 602.

## 15. A borrower-friendly bankruptcy court — Tirole Exercise 4.9, pp. 190–191

**What is asked.** (i) Argue that investors should get the short-term profit and the liquidation value, that the entrepreneur's continuation reward should be exactly $B/\Delta p$, and explain assumptions (1)

and (2). (ii) Write the borrower's optimisation problem and the equation for the continuation cutoff $r^*$. (iii) Show that short-term debt $d = r^*$ implements the contract (default = liquidation) and how $d$ depends on her cash $A$. (iv) A court splits the liquidation proceeds equally with the borrower; show that if $r^* > \hat{r}$ she loses access to financing. (v) If $r^* < \hat{r}$, show that the court either blocks financing or makes bankruptcy more likely, hurting the borrower but not lenders.

**Approach.** Lecture Topic 4B, bankruptcy design. The fixed-investment model with an intermediate verifiable profit $r$ and a liquidation option. Everything not subject to moral hazard ($r$, $L$) is *pledgeable income* at no incentive cost; continuation income is pledgeable only up to $\rho_0(r)$. When the project cannot be financed with continuation in every state, the borrower buys funding by accepting inefficient liquidation where it raises pledgeable income most cheaply, which gives a cutoff rule. We write the Lagrangian, maximise it state by state, and use the investors' breakeven condition to find $r^*$; the court changes investors' payoff from liquidation from $L$ to $L/2$.

**Notation.**

- $r \in [0, \bar{r}]$: verifiable short-term profit, density $f > 0$, c.d.f. $F$; $L > 0$: liquidation value.
- $p_H(r)$, $p_L(r) = p_H(r) - \Delta p$: success probabilities ($p'_H > 0$); $R$: final profit in success; $B$: private benefit.
- $\rho_1(r) = p_H(r)R$: continuation value; $\rho_0(r) = p_H(r)(R - B/\Delta p)$: pledgeable continuation value; both increasing in $r$.
- $\ell(r) \in \{0, 1\}$: liquidation policy; $R_b(r)$: reward on continuation and success; $r^*$: optimal cutoff; $d$: short-term debt.
- $\hat{r}$: $\rho_0(\hat{r}) = L/2$; $r^{**}$: cutoff under the borrower-friendly court.
- $P(c)$, $P_c(c)$: investors' expected income with cutoff $c$ under an enforcing / borrower-friendly court.

**Assumption.** The court's equal split of $L$ cannot be undone by contract.

**Step 1 — (i) What goes to investors; $R_b = B/\Delta p$; conditions (1)–(2).** $r$ and $L$ are not subject to moral hazard, so giving them to investors raises pledgeable income without hurting incentives. On continuation, ($\text{IC}_b$) is $[p_H(r) - p_L(r)]R_b \geq B$, i.e. $R_b \geq B/\Delta p$; any excess is wasted pledgeable income, which would force more liquidation (proved in Step 2). Condition (1), $\rho_1(r) > L$, says continuation is always efficient. Condition (2): always liquidating raises enough, $E[r]+L > I-A$, but always continuing does not, $E[r+\rho_0(r)] < I-A$. Financing therefore needs inefficient liquidation in some states.

**Answer (i).** $r$ and $L$ go to investors because pledging them costs no incentives; $R_b = B/\Delta p$ because any larger reward wastes pledgeable income. (1): continuation is always efficient. (2): liquidating always would raise enough funds, continuing always would not, so some inefficient liquidation is the price of financing.

**Step 2 — (ii) Programme and the cutoff.** With binding breakeven the borrower's utility is the NPV:

$$\max_{\ell(\cdot), R_b(\cdot)} \; E[r] + \int_0^{\bar{r}} \big[\ell(r)L + (1-\ell(r))\rho_1(r)\big] f(r)\, dr - I \tag{97}$$

$$\text{s.t.} \;\; E[r] + \int_0^{\bar{r}} \big[\ell(r)L + (1-\ell(r))p_H(r)(R - R_b(r))\big] f(r)\, dr \;\geq\; I - A, \tag{98}$$

$$R_b(r) \geq B/\Delta p. \tag{99}$$

Let $\mu$ be the multiplier of (98). $\mu > 0$: the unconstrained optimum, $\ell \equiv 0$ by (1), violates (98) by (2). $R_b$ does not enter (97) and lowers (98), so $R_b(r) = B/\Delta p$ and the investors' continuation income is $p_H(r)(R - B/\Delta p) = \rho_0(r)$. The Lagrangian is then

$$\mathcal{L} = (1+\mu)E[r] - I - \mu(I - A) + \int_0^{\bar{r}} \Big\{\ell(r)\,(1+\mu)L + (1-\ell(r))\,[\rho_1(r) + \mu\rho_0(r)]\Big\} f(r)\, dr,$$

which is linear in $\ell(r)$ state by state. Maximising the integrand pointwise: liquidate iff $(1+\mu)L > \rho_1(r) + \mu\rho_0(r)$, i.e.

$$\mu[L - \rho_0(r)] \;>\; \rho_1(r) - L. \tag{100}$$

The left-hand side decreases and the right-hand side increases in $r$, so the policy is a cutoff: liquidate iff $r < r^*$. At the cutoff (100) holds with equality and $\rho_1(r^*) > L$, so $\mu[L - \rho_0(r^*)] = \rho_1(r^*) - L > 0$ and, with $\mu > 0$, $\rho_0(r^*) < L$.

**Step 3 — (ii) The equation for $r^*$.** With cutoff $c$, investors receive

$$P(c) = E[r] + L\,F(c) + \int_c^{\bar r} \rho_0(r) f(r)\,dr, \tag{101}$$

and the NPV, $E[r] + LF(c) + \int_c^{\bar r} \rho_1 f\,dr - I$, falls with $c$ by (1). So $r^*$ is the smallest root of

$$E[r] + L\,F(r^*) + \int_{r^*}^{\bar r} \rho_0(r) f(r)\,dr = I - A. \tag{102}$$

By (2), $P(0) = E[r + \rho_0(r)] < I - A < E[r] + L = P(\bar r)$, so by continuity $r^* \in (0, \bar r)$ (the statement's "$(0, r)$" should read $(0, \bar r)$).

**Answer (ii).** $\max_{\ell(\cdot), R_b(\cdot)}$ (97) s.t. (98)–(99); continue iff $r \geq r^*$, with $r^*$ the smallest root of $E[r] + LF(r^*) + \int_{r^*}^{\bar r} \rho_0(r) f(r)\,dr = I - A$.

**Step 4 — (iii) Short-term debt and the effect of $A$.** Implementation: short-term debt $d = r^*$ due at date 1, any excess $r - d$ also paid to investors (for example as a payout to outside shareholders), and a long-term claim of $R - B/\Delta p$ on success. If $r \geq d$ the debt is paid and the firm continues; if $r < d$ it defaults, creditors take control and liquidate ("bankruptcy"). They do not renegotiate: for $r < r^*$, $\rho_0(r) < \rho_0(r^*) < L$, and the borrower has no cash to bribe them. By Leibniz's rule applied to (101),

$$\frac{dP}{dc} = L\,f(c) - \rho_0(c) f(c) = [L - \rho_0(c)] f(c).$$

Differentiating (102), $P(r^*) = I - A$, with respect to $A$:

$$\begin{aligned} [L - \rho_0(r^*)] f(r^*)\,dr^* &= -dA \\ \frac{dr^*}{dA} &= -\frac{1}{[L - \rho_0(r^*)] f(r^*)} \;< 0. \end{aligned} \tag{103}$$

Short-term debt falls with initial equity: a weaker balance sheet needs more pledgeable income, so the firm must accept bankruptcy in more states.

**Answer (iii).** Short-term debt $d = r^*$ (default and liquidation iff $r < d$), plus a long-term claim $R - B/\Delta p$ on success, implements the optimum; $dd/dA = dr^*/dA = -1/\{[L - \rho_0(r^*)] f(r^*)\} < 0$.

**Step 5 — (iv) Borrower-friendly court with $r^* > \hat r$.** Liquidation now yields investors only $L/2$. For any policy, investors' income in state $r$ (besides $r$) is at most $\max\{L/2, \rho_0(r)\}$. Compare with the enforcing-court contract state by state. For $r < r^*$, $\max\{L/2, \rho_0(r)\} < L$, since $L/2 < L$ and $\rho_0(r) < \rho_0(r^*) < L$. For $r \geq r^* > \hat r$, $\rho_0(r) > \rho_0(\hat r) = L/2$, so the maximum is $\rho_0(r)$, the same as before. Hence

$$E[r] + \int_0^{\bar r} \max\left\{\frac{L}{2}, \rho_0(r)\right\} f(r)\,dr \;<\; E[r] + LF(r^*) + \int_{r^*}^{\bar r} \rho_0 f\,dr = I - A, \tag{104}$$

using (102). No policy lets investors break even: the borrower loses access to financing.

**Answer (iv).** With $r^* > \hat r$, the investors' maximal expected income under the court is below $I - A$: no contract is financed.

**Step 6 — (v) Court with $r^* < \hat{r}$.** Cutoff policies are again optimal (same argument as (100) with $L/2$ for investors). Investors' income with cutoff $c$ is

$$P_c(c) = E[r] + \frac{L}{2}F(c) + \int_c^{\bar{r}} \rho_0(r) f(r)\, dr, \qquad P_c'(c) = \Big[\frac{L}{2} - \rho_0(c)\Big] f(c), \tag{105}$$

which rises up to $\hat{r}$ and falls after it, so its maximum is $P_c(\hat{r})$. If $P_c(\hat{r}) < I - A$, there is no financing. Otherwise the borrower picks the smallest root $r^{**}$ of $P_c(c) = I - A$. For $0 < c \le r^*$, $P_c(c) < P(c) \le I - A$ by (101) and (102), so $r^{**} \in (r^*, \hat{r}]$: bankruptcy is more likely. Lenders break even in either regime, so they are not hurt. The borrower gets the NPV, which falls with the cutoff (Step 3), so $NPV(r^{**}) < NPV(r^*)$: she is hurt. More directly, with an enforcing court she could have written the $L/2$ split into her contract; the friendly court only removes options.

**Answer (v).** If $P_c(\hat{r}) < I - A$ there is no financing; otherwise the cutoff rises to $r^{**} \in (r^*, \hat{r}]$ (more bankruptcy). Lenders still break even; the borrower's utility, the NPV, falls.

**Step 7 — Numerical illustration.** $r \sim U[0,1]$, $p_H(r) = 0.5 + 0.4r$, $\Delta p = 0.2$, $R = 10$, $B = 1.2$ ($R - B/\Delta p = 4$), $L = 4.5$. Then $\rho_1(r) = 5 + 4r > L$, $\rho_0(r) = 2 + 1.6r$, $E[r] = 0.5$, $P(0) = E[r + \rho_0] = 3.3$, $P(1) = E[r] + L = 5$, so (2) holds for $I - A \in (3.3, 5)$; $\hat{r} = (2.25 - 2)/1.6 = 0.15625$, and the court's maximal income is $P_c(\hat{r}) = 3.319531$. With $f = 1$ on $[0,1]$, (101) and (105) integrate to

$$P(c) = 0.5 + 4.5c + 2(1 - c) + 0.8(1 - c^2) = 3.3 + 2.5c - 0.8c^2,$$
$$P_c(c) = 0.5 + 2.25c + 2(1 - c) + 0.8(1 - c^2) = 3.3 + 0.25c - 0.8c^2,$$

so $P_c(\hat{r}) = 3.3 + 0.25(0.15625) - 0.8(0.15625)^2 = 3.3 + 0.0390625 - 0.0195313 = 3.319531$. (iv) $I - A = 4$: $P(r^*) = 4$ gives $0.8r^{*2} - 2.5r^* + 0.7 = 0$, smallest root $r^* = (2.5 - \sqrt{6.25 - 2.24})/1.6 = 0.3109 > \hat{r}$; $\rho_0(r^*) = 2 + 1.6(0.3109) = 2.4975 < L$; and $3.319531 < 4$: no financing. (v) $I - A = 3.31$: $2.5r^* - 0.8r^{*2} = 0.01$ gives $r^* = 0.0040 < \hat{r}$; under the court $0.25r^{**} - 0.8r^{**2} = 0.01$ gives $r^{**} = (0.25 - \sqrt{0.0625 - 0.032})/1.6 = 0.0471$, and the NPV loss is $\int_{r^*}^{r^{**}} (\rho_1 - L)\, dr = \int_{r^*}^{r^{**}} (0.5 + 4r)\, dr = 0.0260$.

**Intuition.** Liquidation is the investors' collateral: it is what they get when the firm cannot pay. A court that hands half of it to the borrower lowers the value of default to creditors. To keep them whole the contract must liquidate more often, which destroys value, or financing breaks down. Lenders still break even, so the whole cost of the borrower-friendly rule falls on the borrower it was meant to protect.

**Common mistakes.**

- Liquidating whenever $\rho_0(r) < L$. Liquidation is always inefficient here (1); it is used only as far as needed to break even, so the cutoff comes from the binding breakeven, not from $\rho_0 = L$.
- Concluding that a court that hands the borrower half of $L$ helps her. Lenders must still break even, so the borrower pays for the transfer through more liquidation or no financing.
- Signing $dr^*/dA$ wrong: more cash means less pledgeable income is needed, so fewer liquidations and *less* short-term debt.

**Book answer.** Matches the book's answer (pp. 586–587): the reasons for pledging $r$ and $L$, $R_b = B/\Delta p$, the programme and $r^*$ as the lowest root of breakeven with $L \ge \rho_0(r^*)$, the debt implementation $d = r^*$, $dr^*/dA < 0$, and (iv)–(v) as read from the book's Figures 2–3 (bankruptcy on $[0, r^{**}]$ with $r^* < r^{**} \le \hat{r}$, lenders not hurt, borrower hurt). The statement's "$(0, r)$" is a typo for $(0, \bar{r})$.

**Ref:** Tirole §3.2.2, pp. 116–117; Exercise 4.9, pp. 190–191; answer pp. 586–587.